Whether Small Transformers Copy a Feature from Context Depends on the Noun's Training History

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Abstract

A language model can predict a word's properties by copying them from context or recalling what it stored. We ask what in a word's training history decides which. We trained 220 small transformers on an artificial language about objects, controlling how often new nouns occurred, with which shapes, and whether their sentences came one at a time or in bursts. Never-seen nouns copied a shape from the preceding sentence, mainly because their input embeddings were untrained. In corpora where object kinds keep their shapes, nouns met one sentence at a time lost most of their copying with exposure; a stored shape lowered copying slightly but outweighed the copied shape. Nouns met in bursts that shared a shape, with the shape changing between bursts, kept copying. In a pre-registered replication with new seeds, after 1,000 sentences they copied 3.7 nats more than bursts with a fixed shape, and 5.5 nats more when every new noun in the corpus had the same history. This copying was not tied to the noun, was carried by its embedding beyond what the embedding's geometry predicts, and partly spread to other nouns. **[One sentence on the 85.6M models.]** In Pythia, replacing nouns' embeddings had no consistent effect.

1 Introduction

When a language model reads *The car was red. The car was*, it can predict *red* because the context says so, or predict a typical car colour from what it has stored. Which source it relies on matters for how models handle documents that contradict their knowledge (Longpre et al., 2021; Xie et al., 2024), for how they learn new words and tokens, and for how we interpret probes that use invented words. Work on in-context and in-weights learning shows that the balance depends on the training data. Classes that occur in bursts, or whose meaning changes, favour learning from context; common, stable classes favour learning stored in the weights (Chan et al., 2022; Chan et al., 2025). Inside pretrained models, mechanisms that copy from the context compete with mechanisms that recall stored facts (Yu et al., 2023; Jin et al., 2024; Ortu et al., 2024). These results describe populations of items. We ask the same question about a single word: *what in a word's own training history decides whether a model copies its features from the context?*

Three accounts are plausible, and they make different predictions.

- **Memory competition.** A model copies until it has stored something about the word; stored knowledge then competes with the context. Words with a stored feature should stop copying sooner than words without one.
- **Representation.** Copying for new words reflects their unusual, untrained embeddings. Once a word's embedding looks like those of familiar words, copying should stop, whatever the word was seen with.
- **Useful context.** A model learns, word by word, whether copying has paid off. A word should keep being copied if, in training, its own earlier mentions predicted its features and memory could not.

Natural data cannot separate these, because frequent words are both familiar and well known. Invented (nonce) words, the usual probe (Thrush et al., 2020; Misra et al., 2023), are new in both respects at once, and conclusions drawn with them can depend on how the new word is represented (Kim et al., 2022; Cacioli, 2026).

We therefore built the separation into training. Small transformers learned an artificial language describing objects by shape, colour and texture, into which we inserted 40 new nouns. For each noun we set how often it occurred, which shapes it occurred with, and whether its sentences came

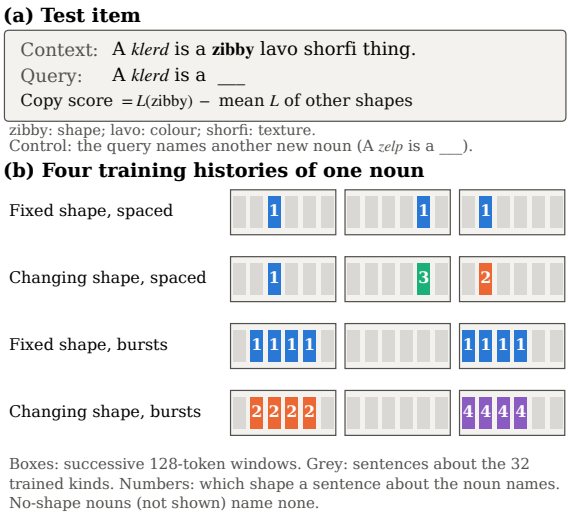


Figure 1: (a) A test item and the copy score, which leaves out the noun's own shape. (b) Four of the training histories, drawn schematically over three training windows; with the same number of sentences, bursts occupy a quarter as many windows as spaced sentences.

one at a time or in short bursts (Figure 1). We measured how strongly each model copied a shape named in one sentence into the next mention of a noun. A first pre-registered experiment tested the memory and representation accounts. A second removed two differences between new nouns and the language's own nouns that the first left open, and tested the useful-context account. A third, registered after the second's results, repeated its key comparisons with new seeds, in a model three times larger, and in corpora where all new nouns share one history. Last, we asked whether the effect of a word's embedding carries over to pretrained models.

In corpora where object kinds keep their shapes, the results favour a combination: copying starts high because new nouns' embeddings are untrained, and is retained only where the noun's own context was useful. Nouns met one sentence at a time lost most of their copying, and a stored shape lowered it only slightly. Nouns whose earlier mentions had been the sole source of their current shape kept copying. What they kept was a tendency, carried by the noun's embedding, to copy the latest shape, not a link between the noun and its shape, and part of it reached other nouns. When the corpus as a whole made shapes changeable, bursts of any kind kept copying. Our claims are about small transformers trained on this language; we note parallels with children's word learning in Section 5.

2 Related work

In-context and in-weights learning. In-context learning emerges when items occur in bursts, more when their meanings change, and trades off against learning in the weights (Chan et al., 2022); it depends on task diversity (Raventós et al., 2023), favours rare classes and can fade as stored solutions take over (Chan et al., 2025; Singh et al., 2023). Its onset has been traced to induction heads (Olsson et al., 2022; Reddy, 2024), and in small transformers stored bigram statistics are learned quickly and in-context ones later, through an induction head (Bietti et al., 2023). Resetting embeddings during training keeps in-context use alive for rare tokens (Anand et al., 2025), and in synthetic biographies frequent entities shift from context to stored knowledge (Kim et al., 2026). We manipulate frequency, stored knowledge and arrangement separately for single words within one model.

Copying and recall. Language models have long gained from copying recent words, through caches and pointers (Kuhn and De Mori, 1990; Grave et al., 2017; Merity et al., 2017), because words recur in bursts: a word that has occurred once is far more likely to occur again (Church and Gale, 1995; Church, 2000), a regularity that human memory also tracks (Anderson and Schooler, 1991). Pretrained models answer from memory more for frequent entities (Mallen et al., 2023; Kandpal et al., 2023; Razeghi et al., 2022) and resist contexts that contradict what they know (Longpre et al., 2021; Xie et al., 2024; Du et al., 2024); copying and recall compete inside the network (Yu et al., 2023; Jin et al., 2024; Ortu et al., 2024).

New words and new tokens. Rare and new words have poor representations (Schick and Schütze, 2020), under-trained tokens behave abnormally (Land and Bartolo, 2024), and new embeddings are often initialised at the mean of existing ones (Hewitt, 2021). Lexical exposure can inflate apparent generalisation (Kim et al., 2022), and when language models acquire a word depends mainly on its frequency (Chang and Bergen, 2022). Children's word learning offers related evidence on shape (Landau et al., 1988; Smith et al., 2002), on expectations about how features vary (Kemp et al., 2007; Dewar and Xu, 2010) and on the timing of exposures (Childers and Tomasello, 2002; Schwab and Lew-Williams, 2016), to which we return in Section 5.

3 Setup

Language. Each corpus describes 32 kinds of object, each named by an invented noun (the *trained kinds*), with 14 exemplars per kind. An exemplar has a shape, a colour and a texture, each one of ten invented words, and is described in one of five sentence frames, such as *A wug is a zibby lavo shorfi thing* (shape, colour, texture) or *Look at the wug. It is zibby and lavo* (Appendix E). The corpus runs through all exemplars many times, each pass arranged afresh, and a model reads the corpus once. In **stable-shape** corpora each kind keeps one shape, so a kind's shape can be stored. In **shifting-shape** corpora a kind gets a new shape in every pass and keeps it within the pass, so its current shape can come only from the context. In both, sentences about a kind come in runs of four; a control corpus uses random order. Every word is one token.

Models. Decoder-only transformers with pre-layer normalisation and learned position embeddings, with 3.4M, 10.9M and 25.6M parameters (4 to 8 layers), and in Experiment 3 also 85.6M (12 layers), read the text once in non-overlapping windows of 128 tokens. Input and output embeddings are untied, so a noun's input embedding (its *row*) carries everything the model has learned about the noun as an input (Appendix C).

Controlling a noun's history. We added 40 new nouns to each corpus. Four were never used in training (*unseen*); the others are *familiar* in the sense that they occur in training. Each familiar noun had one history (Table 1) at a set number of exposures, counted in sentences, with three nouns per condition, rotated over nouns across seeds. A noun's sentences were spread evenly from 5% to 95% of training, and each lay wholly inside one window. Presented *spaced*, a noun has one sentence per window. Presented in *bursts*, it has four consecutive sentences in one window, so the same number of sentences occupies a quarter as many windows.

Three experiments. Experiment 1 (90 models) crossed fixed-, changing- and no-shape nouns, all spaced, with 10 to 10,000 exposures, in stable-shape, shifting-shape and random-order corpora. In it familiar nouns never recurred within a window, unlike the trained kinds, and no-shape nouns never appeared in the frame used for testing. Experiment 2 (60 models) removed both differences. It crossed the three histories with spaced and burst presentation at 100 and 1,000 exposures, in stable-shape and shifting-shape corpora. Fixed- and changing-shape nouns appeared only in the test frame, and no-shape nouns in that frame without the shape (*A noun is a colour texture thing*). This frame has a side effect: no-shape nouns learn to expect a colour word, not a shape, after *is a*, by 1 to 6 nats at the test position, whereas every other familiar noun expects a shape (Appendix I), so comparisons involving no-shape nouns in Experiments 2 and 3 are confounded with expected word class. Experiment 3 (70 models) repeated the stable-shape design of Experiment 2 with ten new seeds at the three sizes and at 85.6M parameters, and trained five seeds at each of the three sizes on **single-history** corpora, in which all 36 familiar nouns had changing shapes or all had fixed shapes. New nouns fill about 60% of training windows in Experiment 1 and 8% in Experiments 2 and 3, so we compare conditions only within an experiment.

History	Sentences about the noun in training
Unseen	none: the noun's embedding is untrained
Fixed shape	always the noun's own shape, which it can store
Changing shape	a new shape every sentence (spaced) or every burst (bursts), so nothing can be stored
No shape	no shape at all: the noun is familiar but has no shape to store

Table 1: Noun histories. In changing-shape bursts the shape is shared within a burst and changes between bursts, so the noun's own earlier sentence is the only source of its current shape.

Measures. A test item gives one sentence about a noun naming a shape s, then the start of the next mention, *A noun is a*, and reads the model's log probability L of each of the ten shape words at the next position (Figure 1a). The **copy score** is $L(s)$ minus the mean of L over the other shapes, leaving out the noun's own shape s^* (a fixed placebo shape for nouns that have none), which s never is. It is in nats: 0 means the context's shape is no more likely than the others, and 3.4 means it is about 30 times as likely as the (geometric) average other shape. Each noun gets 36 items, four for each shape other than its own, with colours and textures drawn at random. Because the copy score leaves out s^*, we also report which shape the model prefers, $L(s^*) - L(s)$. In a **control**, the query names another new noun with the same history; if copying is tied to the noun, the control should score lower. **Knowledge** is $L(s^*)$ minus the mean L of the other nine shapes after an unrelated sentence. We also score unseen

Comparison	Mem.	Rep.	Useful	Exp.
Unseen: typical − untrained row		<0		–
Spaced: fixed − changing shape	<0	0	0	3
Changing shape: bursts − spaced	0	0[b]	>0	2, 3
Bursts: changing − fixed shape	>0	0	>0	3

Table 2: Predictions of the memory, representation and useful-context accounts for the copy score. Blank: no prediction. [b]Or >0 if bursts, which update a noun's row in fewer windows, leave it less trained. Exp.: the experiments in which the comparison was a pre-registered test.

nouns with a **typical embedding**: the mean of the trained kinds' rows plus a random offset whose length is the trained rows' average distance from that mean.

Predictions. Table 2 gives what each account predicts for the comparisons that separate them. The memory account alone predicts that a stored shape lowers copying. The useful-context account alone predicts that changing-shape bursts copy more than changing-shape spaced nouns, unless bursts also keep rows less trained, which the representation account would then predict too. Comparing changing- with fixed-shape bursts, which have the same number of updates, separates useful context from the representation account.

Analysis and registration. The seed is the unit of replication: we average each condition over model sizes within a seed and test across seeds, with Holm correction within each experiment (Holm, 1979), and count a prediction as supported only if its direction also holds in two of the three sizes. Each experiment was registered before any of its models was trained (Experiment 1 before its corpora were built; Experiment 3 after the results of Experiment 2, to test them on new models). Where Experiment 3 tested a finding of Experiment 2, we report the Experiment 3 value, with Experiment 2 as the discovery sample. Each result is labelled pre-registered or exploratory; Appendix G lists every registered test and its outcome.

4 Results

4.1 New nouns copy because their embeddings are untrained

In stable-shape models of Experiment 1, unseen nouns had a copy score of 4.53 nats [95% CI 3.54, 5.52]. Given a typical embedding, the same nouns scored −0.33 (Figure 2, left; exploratory); in Experiment 3 the drop was from 6.88 to 0.79. To see which property of an untrained row matters, we changed only the rows of unseen nouns in the Experiment 1 models. Scaling a row to the trained kinds' average size, keeping its direction, lowered the score only from 4.53 to 4.00 (pre-registered; almost entirely in the smallest models). A random direction of that size kept copying (4.09), whereas the trained kinds' mean row removed it (−0.44; both exploratory). What suppresses copying is therefore what a row shares with trained nouns' rows, not its size. In these small models, then, a probe with an unseen word measures the word's untrained row as much as anything learned about context.

4.2 Nouns met one at a time lose copying

Experiment 1 tested the memory and representation accounts. Changing-shape nouns, which never have a shape to store, are the cleanest test of the memory account. In stable-shape models they lost copying almost as quickly as fixed-shape nouns, which did store their shape (Figure 2, left). Fixed-shape nouns learned their shape well, preferring it by 8.3 to 9.5 nats after an unrelated sentence at 1,000 exposures (pre-registered manipulation check), yet their copy score did not fall faster than that of no-shape nouns (pre-registered prediction, not supported). For all three histories, copying fell from 3.6–4.5 nats at 10 exposures to under 1 nat at 100 and to near zero from 1,000 on. No-shape nouns at 1,000 and 10,000 exposures ended 0.33 nats [0.13, 0.52] above unseen nouns with a typical embedding, within the registered 1-nat equivalence bound (pre-registered). In random-order corpora no noun copied (exploratory), so copying is learned from the runs of sentences about a kind. These results fit the representation account: once a noun was familiar, it lost copying largely whatever it had stored. Experiments 2 and 3 asked whether recurrence, or a context that is useful, keeps copying alive.

4.3 Bursts with changing shapes keep copying

The useful-context account predicts that changing-shape bursts, where a noun's previous sentence gives its current shape and memory cannot, keep copying (Table 2); the other two accounts predict no difference from spaced presentation. Changing-shape bursts scored 4.14 nats [3.60, 4.69] more

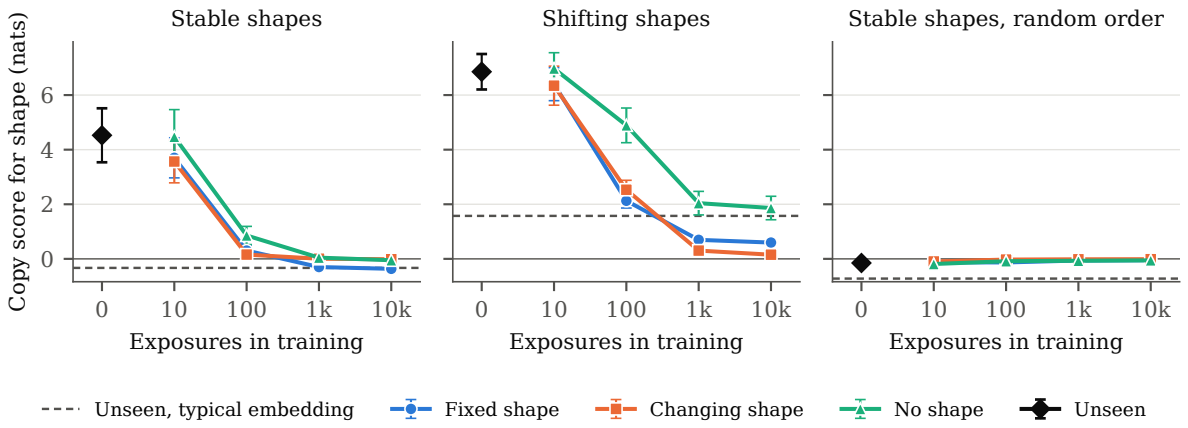


Figure 2: Experiment 1: copy score by number of exposures and noun history, all spaced, in stable-shape, shifting-shape and random-order corpora (means over ten seeds, sizes averaged; 95% intervals). Diamond at 0: unseen nouns. Dashed line: unseen nouns with a typical embedding.

than changing-shape spaced nouns (pre-registered in Experiment 3; Experiment 2: 3.39; Figure 3a; Table 3), as much at 1,000 exposures as at 100. After 1,000 exposures they still scored 4.71 nats, against at most 1.1 for every other familiar history and 0.79 for unseen nouns with a typical embedding. Over training this copying did not fade: in Experiment 2, for nouns with 1,000 exposures, changing-shape bursts copied 0.8 to 2.6 nats at the first saved weights, depending on model size, and 3.85 at the end, while every other history stayed below 2 nats and ended near or below 1 (exploratory; Figure I4).

Because bursts occupy fewer windows, a burst noun's row is updated less often, which might leave it closer to an untrained row. Fixed-shape bursts control for this: the same frame, presentation and number of windows, but a shape that never changes, so memory can supply it. At 1,000 exposures changing-shape bursts copied 3.68 nats [2.99, 4.37] more than fixed-shape bursts, at all three sizes (pre-registered in Experiment 3; found in Experiment 2 at 2.86, exploratory). This comparison separates useful context from the representation account (Table 2).

Recurrence without a useful context helped less and only early. No-shape bursts scored 0.80 nats [0.47, 1.13] more than no-shape spaced nouns in Experiment 2 (pre-registered), but the gain was confined to 100 exposures: at 1,000 it was 0.05 [−0.30, 0.40] (Experiment 3: 0.20 [−0.10, 0.49]; exploratory). Fixed-shape bursts also copied more than fixed-shape spaced nouns, but at 100 exposures bursts left the stored shape weaker (knowledge 3.0 against 5.4 nats), so the registered test of this comparison could not be interpreted. At 1,000 exposures, where knowledge was equal, bursts added about 0.5 nats (0.47 and 0.57 in the two experiments; exploratory), a sixth to a seventh of the advantage of changing-shape bursts.

	Exp. 2		Exp. 3	
Comparison	Pooled	1,000	Pooled	1,000
Changing: bursts − spaced	3.39^{r}	3.35	4.14^{r}	4.15
Bursts: changing − fixed	1.73	2.86	2.52	3.68^{r}
No shape: bursts − spaced	0.80^{r}	0.05	0.69	0.20
Spaced: fixed − changing	−0.02	0.02	-0.25^{r}	−0.10
Spaced: fixed − no shape^{a}	-1.02^{r}	−0.46	−1.00	−0.39
Same noun − other noun	0.05	0.05	0.07	0.08^{r}

Table 3: Key comparisons in stable-shape corpora (copy score, nats; seed-level means; 100 and 1,000 exposures pooled, and 1,000 alone). ^{r}Pre-registered test; all other values exploratory. Same noun − other noun: the control, for changing-shape bursts. ^{a}Confounded with expected word class.

4.4 A stored shape lowers copying slightly but wins in prediction

Fixed- and changing-shape nouns presented spaced share frame, presentation and expected word class, and differ only in whether one shape can be stored. Fixed-shape spaced nouns copied 0.25 nats [0.15, 0.35] less (pre-registered in Experiment 3), mostly at 100 exposures (0.40) rather than at 1,000 (0.10). The difference was reliable but not shown to be smaller than 0.3 nats, so a stored shape did lower copying; Experiment 1 gave a similar 0.30 at 1,000 exposures, and Experiment 2 none (−0.02,

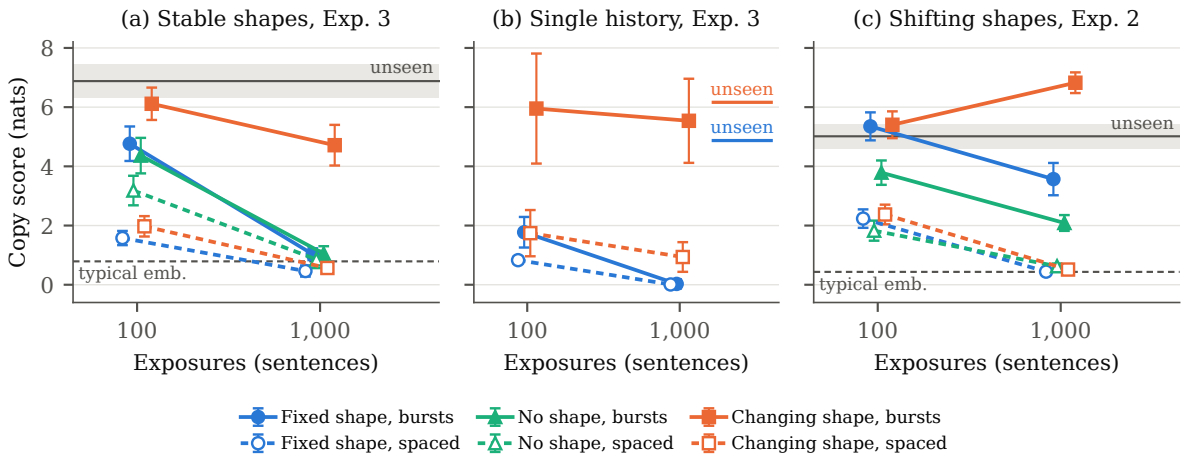


Figure 3: Copy score by history (colour) and presentation (solid: bursts; dashed: spaced) at 100 and 1,000 exposures (means over seeds, sizes averaged; 95% intervals; points offset for legibility). (a) Stable-shape corpora, Experiment 3 (ten new seeds). (b) Single-history corpora, Experiment 3 (five seeds): all familiar nouns have changing shapes (orange) or all have fixed shapes (blue). (c) Shifting-shape corpora, Experiment 2 (exploratory). Solid grey line and band: unseen nouns; dashed line: unseen nouns with a typical embedding. Experiment 2's stable-shape results are in Appendix I.

equivalent to zero within ±0.3; TOST Lakens, 2017). Which shape the model finally predicts is another matter. Fixed-shape spaced nouns preferred their own shape to the context's by 5.84 nats [5.57, 6.11] (pre-registered), 4.1 at 100 exposures and 7.6 at 1,000. A stored shape thus barely suppressed the copying itself but outweighed it in prediction. The registered comparisons with no-shape nouns suggest a larger effect: fixed-shape spaced nouns scored about 1 nat below no-shape spaced nouns in both experiments. But no-shape nouns expect a colour where the others expect a shape, so these differences reflect expected word class, not a stored shape.

4.5 History sets whether the model copies, not what it copies

If changing-shape bursts taught the model to bind a noun to its current shape, the control, in which the query names another noun, should score lower. It did not. For changing-shape bursts at 1,000 exposures the context about the queried noun added 0.08 nats [0.02, 0.14] over the control, within ±0.3 (pre-registered), and no stable-shape condition in any experiment exceeded 0.14. The model copied the most recent shape after the queried noun, whichever noun the preceding sentence described. To see whose history matters, we paired a context about a noun of one history with a query naming a noun of another (Table J3). At 1,000 exposures, a queried noun with changing-shape bursts copied 1.72 nats [1.36, 2.08] more than one with fixed-shape bursts, averaged over the context noun's history, whereas the context noun's history added 0.47 [0.36, 0.58]; the difference, 1.25 [0.94, 1.56], was pre-registered in Experiment 3 (Experiment 2: 1.35 against 0.39). In training, every sentence of a burst followed one about the same noun, so nothing required the model to check which noun a shape belonged to. A registered search for induction-like heads (Olsson et al., 2022) in Experiment 1 found none: no head matched prefixes clearly better than uniform attention, and ablating the best candidates removed no more copying than random heads (Appendix H). The mechanism remains open.

4.6 Copying spreads between nouns

Because copying is not tied to the noun, what changing-shape bursts teach may reach other nouns in the same corpus. In Experiment 3's single-history corpora no noun of another history was present (Figure 3b). There, changing-shape bursts copied 5.51 nats [4.24, 6.78] more than fixed-shape bursts at 1,000 exposures, at all three sizes (pre-registered), so the main difference does not need nouns of the other history. It was larger than in mixed corpora because fixed-shape bursts lost all their copying when no changing-shape noun shared the corpus: 0.03 nats, against 1.03 in mixed corpora with the same seeds, a difference of 1.00 [0.80, 1.20] (pre-registered). For changing-shape bursts, the corresponding comparison was incon-

clusive (+0.24 [−1.44, 1.92]). Some of the copying that other histories keep in mixed corpora is therefore learned from changing-shape nouns.

4.7 The copying lives in the noun's row

A noun enters the model only through its row, so whatever its history taught must be carried there. Could changing-shape bursts simply leave the row less trained? For every familiar noun we measured the row's cosine with its initial value, its distance from the trained kinds' mean row and its norm, and regressed copy scores on them within each model. At 1,000 exposures changing-shape bursts copied 3.67 nats [3.03, 4.32] more than these measures predict (pre-registered in Experiment 3; Experiment 2: 3.19, exploratory), although their rows were only slightly closer to their initial values than those of fixed-shape bursts (cosine 0.85 against 0.78 in both experiments). At 100 exposures geometry still accounted for their copying (−0.09 [−0.51, 0.33] in Experiment 2; exploratory). Swapping rows between the two histories moved the copying with the row (Experiment 3: 1.01 and 4.70 nats; exploratory), as it must, since each noun is a single token. Bursts thus first preserve copying by leaving rows near their initial state, and changing-shape bursts then give the row a copying tendency of its own. In Experiment 2, first-layer attention from the query to the context's shape tracked copying across nouns (Spearman $\rho = 0.81$).

This learned copying is narrow (both experiments, exploratory). With one unrelated sentence between the context and the query, it almost vanished (at most 0.44 nats). With the context in another frame (*Look at the noun. It is shape and colour.*), changing-shape bursts kept 35 to 40% of it. For colour, which varies freely within every kind, no noun copied.

4.8 When the whole corpus rewards copying

These conclusions hold for stable-shape corpora. In Experiment 2's shifting-shape corpora, where the trained kinds' own shapes can be recovered only from context, copying behaved differently (Figure 3c; all exploratory). Part of it was tied to the noun: a context about the queried noun raised the copy score over the control by 0.48 nats for unseen nouns and 1.45 for changing-shape bursts at 1,000 exposures. Every burst history kept copying: at 1,000 exposures, fixed-shape bursts scored 3.57 nats and no-shape bursts 2.09, although bursts again weakened stored shapes. And changing-shape bursts copied more than unseen nouns (6.82 against 5.02). What a noun's history teaches therefore depends on what the rest of the corpus teaches: when the corpus rewards copying, recurrence alone sustains it.

4.9 Model size

Most effects were largest in the smallest models (Table I7). In Experiment 3, the advantage of changing- over fixed-shape bursts at 1,000 exposures was 6.36, 2.44 and 2.24 nats at 3.4M, 10.9M and 25.6M parameters, while unseen nouns with a typical embedding copied more as models grew (−0.19, 1.20 and 1.37). **[85.6M models: the registered test (H57) and the cell values.]**

4.10 Pretrained models

Finally, we asked whether the untrained-embedding effect carries over to pretrained models. We registered three predictions for Pythia models from 1B to 12B parameters (Biderman et al., 2023): that replacing a noun's embedding with a random one of untrained scale would raise copying, more than replacing it with a typical one, and, by the memory account, that nouns with a typical colour would copy less than nouns without one. Each needed support in four of the five models. We used 22 pairs of one-token nouns, one colour-typical (*sky*, *leaf*, *bone*) and one colour-variable (*truck*, *hat*, *dog*), matched on frequency in the Pile (Gao et al., 2020; Liu et al., 2024) and on count or mass use, in frames such as *The noun was colour. The noun was*. Because real nouns have colour priors, the copy score here compares contexts: the log probability of a colour after a context naming it minus that after a context naming another, averaged over the 11 basic colour words. The models had stored typical colours: with no context, colour-typical nouns preferred their colour by 1.3 to 1.7 nats, and human colour norms (Paik et al., 2021) agreed with our coding for 15 of the 16 colour-typical nouns they cover (both exploratory).

Real nouns copied 1.9 to 2.7 nats of colour (Figure 4). None of the predictions held (pre-registered). The random replacement raised copying reliably in two models and lowered it in two others, by up to 0.4 nats; the typical replacement changed it by up to 0.6 nats in either direction; and colour-typical and colour-variable nouns did not differ reliably. Most changes were not small enough to count as zero (within ±0.3 nats), so the result is inconsistent rather than null. In the

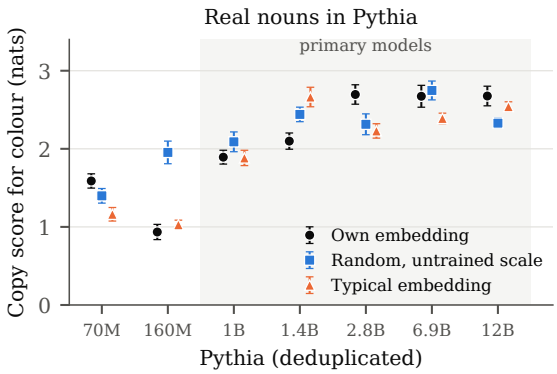


Figure 4: Copy score for colour of real nouns in Pythia models, with their own input embeddings or replaced ones (means over 44 nouns, 95% intervals). The registered tests used the 1B to 12B models; 70M and 160M are descriptive.

70M and 160M models, outside the registered set, the pattern differed again: at 160M parameters the random replacement doubled copying. The untrained-embedding effect of the small models did not transfer to these models and this probe.

5 Discussion

What decides copying. In stable-shape corpora our results fit a combined account. Whether the model copies the latest shape after a noun depends on the noun's row. The tendency starts high for rows far from the trained nouns' rows, which is why unseen nouns copy and lose it with a typical row. Training removes it, largely whatever is stored, unless the noun's own earlier sentences were the only source of its current shape: changing-shape bursts kept copying where fixed-shape bursts with the same recurrence did not, in new seeds and in corpora without nouns of the other history. The memory account found little support. A stored shape lowered copying by 0 to 0.4 nats across three experiments, a tenth of the effect of a useful context, and acted mainly by outweighing the copied shape in prediction. Row analyses support the combined account: early on, bursts keep copying by keeping rows near their initial state; later, changing-shape bursts give the row a copying tendency that its geometry does not explain.

What is expected and what is not. That a model copies where copying lowers its loss is expected; three things are less so. Recurrence alone did not keep copying in stable-shape corpora: no-shape bursts lost their advantage over spaced presentation by 1,000 exposures. A stored shape barely suppressed copying: memory and context added up, and the stored shape won by outweighing the copied one. And the copying was not tied to the noun and spread to other nouns, fitting training in which it never needed to be bound to a noun. Together, this is the population-level finding of Chan et al. (2022), that bursty items whose meaning changes favour learning in context, acting on single words within one model, but only when the rest of the corpus does not already reward copying.

Implications. In small models trained this way, probes with invented words can measure an unseen token's untrained row more than anything learned about context, so effects should be checked with rows that look familiar. Because copying learned for some words spread to others, a word's tendency to copy depends on the company it keeps in training, not only on its own history. Pythia did not show the embedding effect; whether pretrained models' copying depends on a word's history needs measures of each word's burstiness in pretraining.

Analogies with children. Two findings echo research on children's word learning, as analogies only. Changing-shape bursts resemble the boxes of Dewar and Xu (2010), uniform within and varying between; after one draw from a new box, 9-month-olds expected the next to match, an overhypothesis in the sense of Kemp et al. (2007). And at 100 exposures spaced nouns stored their shape more strongly than nouns met in bursts, while bursts kept the context in use. Children likewise retain words and categories better from spaced than from massed exposures (Childers and Tomasello, 2002; Vlach et al., 2008), although they learn new labels better when the labels are repeated in successive sentences (Schwab and Lew-Williams, 2016).

Limitations

The language is artificial, with 32 kinds, three features and five sentence frames; the models are small, trained with one recipe, and most effects shrank with model size. Experiments 2 and 3 tested 100 and 1,000 exposures, and over training exposure is confounded with training time. The learned copying almost vanished beyond one sentence, transferred only partly to another frame, and did not apply to colour. No-shape comparisons are confounded with expected word class, and bursts weakened stored shapes at 100 exposures. Single-history corpora used five seeds, and whether other nouns change the copying of changing-shape bursts

was inconclusive. The unseen-noun result depends on untied embeddings, which leave unseen rows untrained. The head search could not test its hypothesis, and the mechanism is unknown. Planned power for a 1-nat difference in Experiment 1 was 0.57, below the target of 0.8. The pretrained test used one model family, 22 noun pairs, three frames and colour only. Earlier experiments in this language were registered in stages, some after related results were known (Appendix A).

Ethics statement

All training data are artificial; the pretrained models are public. No data were collected from people. Code, corpora, results and all trained models will be released; anonymised copies are provided as supplementary material.

Use of AI assistants

An AI assistant (Claude, Anthropic) was used to help design analyses, write code, review code and design, draft and edit text and check citations. It also reviewed the write-up of Experiment 1, which identified the two differences addressed by Experiment 2, and coded the colour typicality of the pretrained-model nouns in two independent passes. The authors are responsible for all content, including the accuracy of every citation.

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A Registration record

In the registrations, Study 1 is the earlier training experiments (Section 3; Appendices C and E), Study 2 an analysis of natural corpora and Study 3 an earlier test of pretrained models. Studies 2 and 3 do not bear on the claims of this paper; their methods and full results, and further results of Study 1, are in the long version in the supplementary material. Table A4 reports every registered hypothesis of all three studies.

All registrations are in one public project on the Open Science Framework (link withheld for review; an anonymised view-only link is in the supplementary material), and each states what was known when it was written. Table A1 lists them, Table A2 gives the label each registration uses for each row of Table A4, and Table A4 gives every registered hypothesis and its result.

Table A1: Registrations. Each was written before the models it concerns were trained or scored, or before the transcripts were read.

Registration	Written	Rows of Table A4	Known when it was written
Original	2 Mar 2026	Corpora, models and controls (long version)	Nothing trained
Amendment 1	7 Oct 2026	1, 2, 3, 6, 16	Random-order models trained and scored on an earlier test, with the old loader; no grouped model trained
Amendment 2	7 Oct 2026	7 (texture added as a co-primary test)	No grouped model trained
Amendment 3	8 Oct 2026	23, 24, 25	Study 1 results for rows 1 to 3, 6, 7 and 16 (old loader); no pretrained model scored
Amendment 4	8 Oct 2026	20, 21	Studies 1 and 3 results; no transcript read
Amendment 5	9 Oct 2026	4, 5, 8, 9, 17, 18, 19, 22	All earlier results (Study 1 with the old loader); four reviews of a draft; five exploratory checks, including the confound of the noun with another
Amendment 6	11 Oct 2026	11, 12, 13	Amendment 5 results (old loader); a prediction for the new models fitted to them
Amendment 7	12 Oct 2026	10, 14, 15	All results, with corrected scores
Amendment 8	15 Oct 2026	26 to 32	All results, with corrected scores; three rounds of reviews, which noted that fixed-file models memorised their files and that the novel-noun test was uninformative; no fresh-text model trained
Amendment 9	23 Oct 2026	Familiarity study (Appendix H)	All results above; no familiarity corpus built
Amendment 10	24 Oct 2026	Recurrence, frames, rows and pretrained models (Appendix I)	Amendment 9 results and a blind review of them; no Amendment 10 corpus used, model trained or pretrained model run on its items

Table A2: Rows of Table A4 and the labels used in the registrations.

Row	Label	Row	Label	Row	Label	Row	Label
1	H4b	9	H13b	17	H14a	25	H8
2	H1b	10	H22	18	H14b	26	H23
3	H2b	11	H17	19	H15	27	H24
4	H11	12	H18	20	H9	28	H25
5	H12	13	H19	21	H10	29	H26
6	H5b colour	14	H20	22	H16	30	H27
7	H5b texture	15	H21	23	H6	31	H28
8	H13a	16	H3b	24	H7	32	H29

The loading correction. The model code ties the input embedding to the output layer. On the graphics

card used for training (through DirectML) the tie was lost, so every Study 1 model was trained and saved with two separate matrices. The first version of the scoring code rebuilt each model with the tie in place before loading its weights, so the output matrix overwrote the input embedding. We found this on 12 October 2026, fixed the loader, and rescored all 315 Study 1 models then trained, 461 intermediate checkpoints of the random and grouped Regular models and the trained-noun items, with the same items. A check on eight randomly chosen models confirmed the fix: mean loss on each model's own training corpus was 0.07 to 0.16 nats per token with the corrected loader, close to the last logged training loss (0.08 to 0.23; dropout is off at test), and 0.58 to 1.97 with the old one. Studies 2 and 3 were not affected. The old scores are kept in the repository.

Table A3: Study 1 registered tests before and after the correction (shape-over-colour index, nats). Rows 10, 14 and 15 were registered and scored after the correction.

Row	Old loader	Corrected	Verdict before → after
1	1.36 vs 0.18	6.49 vs 0.47	Supported → supported
2	0.18	0.47	Not supported → not supported
3	0.18 vs 0.09	0.47 vs −0.25	Not supported → supported
4	difference 1.45	difference 6.93	Supported → supported
5	difference 1.17	difference 2.76	Supported → supported
6	1.36 vs 0.80	6.49 vs 2.12	Supported → supported
7	1.10 vs −0.04	5.36 vs 0.46	Supported → supported
8	slope 0.29	slope 1.63	Supported → supported
9	0.24 vs 0.18	1.43 vs 0.47	Not supported → supported
11	0.21 vs 0.18	0.50 vs 0.47	Not supported → not supported
12	0.29 vs 0.24	0.80 vs 1.43	Not supported → not supported (reversed)
13	0.21 vs −0.08	0.50 vs 0.10	Supported → supported
16	difference 0.0007	difference 0.005	Equivalent to zero → equivalent to zero

Table A4
All registered hypotheses and results; registered secondary analyses are reported in the text. Rows 1 to 16 concern fixed-file models and rows 26 to 32 fresh-text models. Seed-level tests average the three sizes within each of five seeds (t-based 95% intervals, two-sided; tests one-sided unless marked, 4 degrees of freedom), so a one-sided test can be below .05 while the two-sided interval includes zero (row 13). The registered p is the test named in the registration, one-sided unless marked, Holm-corrected within its registration; for rows 26 to 32 it is the seed-level test, corrected within rows 26 to 29 and within rows 30 to 32. Corpus intervals resample sources and then documents; Pythia and BabyLM intervals resample nouns.

#	Prediction	Result (nats)	Seed-level difference [95% CI]	Seed-level p	Registered p	Verdict
	1. Does grouping teach the fixed feature?					
1	Grouped order gives a higher index than random order	6.49 vs 0.47	6.02 [5.24, 6.79]	$< .001$	$< .001$	Supported
2	Random order gives an index above zero	0.47	0.47 [−0.61, 1.56]	.15	.061	Not supported
3	Random order: Regular above Mixed-slot	0.47 vs −0.25	0.73 [−0.69, 2.15]	.11	.032	Supported by the registered test, not at the seed level
4	Grouped: Regular above No stable feature	6.49 vs −0.44	6.93 [5.47, 8.40]	$< .001$	$< .001$	Supported
5	Grouped: Colour stable below No stable feature	−3.20 vs −0.44	2.76 [1.99, 3.52]	$< .001$	$< .001$	Supported
6	Grouped: Regular above Mixed-slot, shape over colour	6.49 vs 2.12	4.37 [1.98, 6.76]	.004	$< .001$	Supported
7	Grouped: Regular above Mixed-slot, shape over texture	5.36 vs 0.46	4.89 [2.69, 7.10]	.002	$< .001$	Supported
8	The index rises with log B across six orders	Slope 1.63 (SE 0.10)	1.63 [1.46, 1.80]	$< .001$	$< .001$	Supported
9	B = 3.6 above random order	1.43 vs 0.47	0.96 [0.63, 1.28]	$< .001$	$< .001$	Supported
10	The same, Objects corpora, different-object runs	1.18 vs 0.22	0.96 [0.37, 1.54]	.005	$< .001$	Supported
26	Fresh text: grouped order gives a higher index than random order	9.52 vs −1.56	11.08 [9.57, 12.59]	$< .001$	$< .001$	Supported
27	Fresh text: B = 3.5 above random order	1.28 vs −1.56	2.84 [1.41, 4.27]	.003	.005	Supported
28	Fresh text, grouped: Regular above No stable feature	9.52 vs −3.98	13.50 [11.48, 15.52]	$< .001$	$< .001$	Supported
	2. What kind of grouping helps?					
11	512-token window above 128, random order	0.50 vs 0.47	0.03 [−0.44, 0.50]	.44	.87	Not supported
12	512-token window above 128, B = 3.6	0.80 vs 1.43	−0.63 [−0.90, −0.37]	.003 (two-sided)	1.00	Not supported; reliably reversed
13	512-token window: Regular above No stable feature	0.50 vs 0.10	0.40 [−0.04, 0.85]	.032	.045	Supported; weak
14	Grouped: different-object above same-object runs	5.65 vs 2.99	2.66 [1.57, 3.75]	.001	$< .001$	Supported
15	B = 3.6: different-object above same-object runs	1.18 vs 0.45	0.73 [0.15, 1.32]	.013	$< .001$	Supported
29	Fresh text, grouped: different-object above same-object runs	7.60 vs 5.87	1.72 [0.54, 2.90]	.008	.008	Supported
	3. Is the expectation tied to the noun?					
16	Fixed files, random order: index higher after the same noun (novel nouns)	Difference 0.005; equivalent to zero (p = .001), but the novel nouns were untrained (Section 3)			.46	Not supported
30	Shape reassigned, grouped: noun-specific index above zero, novel nouns	0.37	0.37 [0.24, 0.49]	$< .001$	.001	Supported

#	Prediction	Result (nats)	Seed-level difference [95% CI]	Seed-level p	Registered p	Verdict
31	The same, above grouped Regular (fresh text)	0.37 vs 0.12	0.25 [0.10, 0.39]	.005	.005	Supported
32	Shape reassigned, grouped: noun-specific index above zero, nouns from training	11.24	11.24 [10.44, 12.03]	$< .001$	$< .001$	Supported
17	Pythia: noun-specific index rises with size (wording matched)	$\rho = -.42$			1.00	Not supported
18	Pythia 410M to 12B: noun-specific index above zero (wording matched)	−0.15 (SE 0.12)			1.00	Not supported
19	Pythia: 410M above 160M across ten training seeds 4. What do natural input and pretrained models show?	0.15 vs −0.15			.012	Supported
20	Speech to children: B above 1	3.57 [3.13, 4.30]			$< .001$	Supported
21	Speech to children above written text	3.57 vs 1.43; difference 2.14 [1.55, 2.91]			$< .001$	Supported
22	Speech to children differs from adult conversation (two-sided)	3.57 vs 2.33; difference 1.24 [0.51, 2.03]			$< .001$	Supported; higher
23	Pythia: index rises with size	$\rho = .92$			.003	Supported
24	Pythia: index rises over training	Mean within-size $\rho = -.34$			1.00	Not supported
25	BabyLM baselines (10M, 100M words): index above zero	0.47 [−1.05, 1.93]; 0.37 [−0.04, 0.73]			.52; .12	Not supported

Fresh text and shape reassignment. Rows 26 to 32 were registered on 15 October 2026, after reviews of the manuscript had noted that fixed-file models memorised their files and that the test with novel nouns was uninformative, and before any fresh-text model was trained. The corpora were built, and their checksums registered, before training. Amendment 8 names the conditions by their folder names (for example reassigned_grouped).

B Deviations from the registrations

1. Two registered analyses are not reported here: the Bayesian reference model of Amendment 1, which is in the repository, and the prediction of Amendment 6 for the 512-token models, which had been fitted to scores later corrected (Appendix A).

2. Added after registration: the seed-level tests for rows 1 to 3 and 6 to 8, the seed-level intervals, and intervals for the differences between natural corpora (from the registered two-stage bootstrap). Effective clustering was computed before it was registered as the scale for a figure of the long version and is reported there as exploratory.

3. The two models of about 70 billion parameters were scored with Apple's MLX library, because PyTorch crashed the machine; on smaller models the two differed by 0.003 to 0.010 nats on average.

4. In the hand coding of Study 2, a rule for labels not used for a physical object was added during coding and applied by both coders, and one pair of the reliability subset was discussed with an AI assistant and left out of the agreement check.

Smaller technical changes (software versions, download addresses, a stale file in one public model repository) are listed in the repository.

C Model and training settings

Table C1: AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up then cosine decay, dropout 0.1, untied input and output embeddings. Fixed-file models read windows starting at random positions in the corpus; fresh-text models read non-overlapping windows, each at most once, in random order (the 3.4M and 25.6M models used about 60% of the windows available, the 10.9M models 94 to 96%). Byte-pair tokeniser with 351 entries for the Regular corpora and 314 to 339 for the other corpus families; every artificial word is one token. Seeds 42, 123, 456, 789 and 1001 each set the corpus, the initial weights and the batch order. These five seeds are those of the earlier experiments; Experiments 1 and 2 used ten (adding 2718, 3141, 1618, 1414 and 1732), untied embeddings by design, a tokeniser of 329 entries and weights saved every 1,000 steps. All models are pre-LayerNorm decoders with learned absolute position embeddings, GELU feed-forward layers and a final LayerNorm. Checkpoints were saved every 500 steps. Training ran on one Windows PC through DirectML.

Setting	3.4M	10.9M	25.6M
Layers, heads, width	4, 4, 256	6, 8, 384	8, 8, 512
Feed-forward width	1,024	1,536	2,048
Steps	5,000	8,000	10,000
Batch (128-token windows)	32	32	16
Batch (512-token windows)	8	8	4
Peak learning rate	3×10^{-4}	2×10^{-4}	1×10^{-4}
Warm-up steps	200	400	500

Table C2: Memorisation. Mean next-token loss (nats per token) on 300 windows from each model's own training text and from a new arrangement of the same sentences (100 further passes, 27 for the Objects corpora, never used in training); 15 models per cell. At the seed level, the gap between the two losses was larger for fixed-file than for fresh-text models by 0.57 to 1.16 nats, with every interval excluding zero.

Corpus	Fixed file: own file	Fixed file: new arrangement	Fresh text: own text	Fresh text: new arrangement
Regular, random	0.11	1.25	0.55	0.55
Regular, B 3.6 (fresh: 3.5)	0.10	1.26	0.54	0.54
Regular, grouped	0.09	0.85	0.35	0.35
No stable feature, grouped	0.09	1.00	0.39	0.39
Objects, same-object runs, grouped	0.08	0.66	0.28	0.28
Objects, different-object runs, grouped	0.09	0.76	0.32	0.32

D Data and code

All code, corpora, test items, model scores and analysis outputs, and the final weights and training logs of every trained model, will be released on publication; anonymised copies of the code and results are in the supplementary material. A README gives the command that regenerates each table and figure, with MD5 checksums for the fixed inputs. The pretrained models are public at the revisions listed.

E Language, corpora and earlier measures

This appendix describes the artificial language and the measures of the earlier experiments.

E.1 Corpora

Each corpus describes 32 kinds of object, each named by a novel noun (*wug*, *dax*, *blorp* and so on), with 14 exemplars per kind. Each exemplar has a shape, a colour and a texture, each a novel word from a list of ten per dimension, and is described by one sentence in one of five frames:

1. *A wug is a [shape] [colour] [texture] thing.*
2. *Look at the wug. It is [shape] and [colour].*
3. *Here is another wug. This one is [shape].*
4. *The wug I showed you is [shape], but a different [colour].*

5. *That is a wug because it is [shape].*

Shape is named in all five frames, colour in three and texture in one. The frames are the same in every corpus we compare, so they cannot by themselves explain a difference between corpora, although they may interact with what is manipulated. The 448 sentences of a corpus, each used once, make up one pass. Five corpora were generated for each condition, one per seed.

Content. The corpora differ in which feature, if any, stays fixed within a kind (long version, Table 1). In **Regular** corpora every kind keeps one shape across its 14 exemplars, while colour and texture vary. In **Mixed-slot** corpora each kind keeps one word from the shape list, but for about half the kinds that word appears in the texture slot, and colour varies; no single dimension is fixed across all kinds. In **No stable feature** corpora, built from Regular, each exemplar's shape is drawn at random from the list, so nothing is fixed within a kind. In **Colour stable** corpora, also built from Regular, shape is drawn in the same way and each kind keeps one colour instead. Frame 4 is unchanged, so in Colour stable corpora it names the kind's fixed colour after the words *a different*.

Objects. A further set of corpora separates two kinds of nearby repetition. In the Regular corpora, two sentences about a kind always describe different exemplars. In speech, nearby mentions often return to the same object. The Objects corpora keep the Regular content but give each kind 13 objects and describe each object four times, once in each of frames 1, 2, 3 and 5. Frame 4 is left out because it states that the colour differs. A pass of these corpora has 1,664 sentences.

Order. The same sentences were arranged in several orders. In **random** order every pass is shuffled. In **grouped** order, within each pass, the sentences about each kind are shuffled, cut into runs of four, and the runs are shuffled across kinds. Every sentence occurs exactly as often as in random order. In **graded** orders each run is kept with a set probability and otherwise split into single sentences before shuffling, which gives intermediate levels of clustering. In the Objects corpora, a **same-object run** is the four descriptions of one object, and a **different-object run** is four different objects of one kind, one per frame. A Latin square over objects and frames ensures that every sentence is used once per pass in every order. Same-object and different-object runs were each arranged fully grouped and at an intermediate level.

Reassigned. In every corpus so far, a kind's fixed feature is fixed for the whole corpus, so the noun alone predicts it, and a model can predict the shape in a sentence from its noun without comparing the sentence with the one before. In Reassigned corpora, built from Regular, each kind receives a new shape, drawn at random from the ten, in every pass, and keeps it in all of its sentences in that pass. The next sentence about the same kind still repeats the shape, but the noun no longer says which shape it will be. These corpora were arranged in random and grouped order.

Fresh and fixed text. A model trained many times over on one short file can memorise it. Our main models were therefore trained on fresh text: 7,500 passes of a corpus (2,100 for the Objects corpora), each arranged with its own random draw by the rule of its order, 34 to 35 million tokens in all. Training read this text in non-overlapping 128-token windows, so no window was seen twice. A second set of models was trained in the common way, on one fixed file of 11,648 sentences (26 passes, or 7 for the Objects corpora; about 120,000 tokens), with windows starting at random positions, so that each position was seen 160 to 290 times. These fixed-file models memorised their files. Their loss was 0.08 to 0.11 nats per token on their own file but 0.66 to 1.26 on a new arrangement of the same sentences, while fresh-text models scored 0.28 to 0.55 on both (Table C2). Table 1 of the long version shows which conditions were trained on fresh text and which on a fixed file; where both exist, we report fresh text first. The fresh graded order, built by the same rule as the fixed one at B = 3.6, measured B = 3.5.

Clustering. We measure clustering with one statistic, B, here and in Study 2. Take the sequence of mentions of object nouns in a text, and every pair of consecutive mentions that are no more than 100 words apart. P is the share of these pairs that name the same kind. P_0 is the same share after the sentences (or utterances) of each document are shuffled, averaged over shuffles. B = P/P_0, pooled over documents. For example, in *Look at the dog. The dog is big. Where is the cup? Here is another dog.*, the consecutive pairs are dog–dog, dog–cup and cup–dog, so P = 1/3. B = 1 means no clustering beyond what the content predicts. Random-order Regular corpora have B = 0.9, slightly below 1, because each pass is shuffled

separately: within a pass, a sentence's kind has 13 other members among 447 sentences, against 363 among 11,647 when the whole corpus is shuffled. The graded levels (long version, Table 1) were chosen after the clustering of natural input had been measured (Study 2): 1.4 and 3.6 match B in written text (1.43) and in speech to children (3.57). They were fixed from the corpora alone before any graded-order model was trained.

Table E1: Study 1 corpora. Every corpus describes 32 kinds. Fresh: 7,500 passes (2,100 for Objects), each arranged anew and read once in non-overlapping 128-token windows. Fixed: one file of 11,648 sentences (10,304 to 12,992 for three further controls: No kind label, Paraphrased and Weak label), read in 128-token windows at random positions unless marked. Each order of each corpus was trained at three sizes with five seeds (15 models). The further controls are reported in the long version.

Corpus	What stays fixed within a kind	Orders (measured B)	Text	Models
Regular	Shape. Colour and texture vary.	Random (0.9); graded (3.5); grouped (23)	Fresh	45
No stable feature	Nothing	Grouped (23)	Fresh	15
Objects	Shape; each object described four times	Same-object runs (24); different-object runs (24)	Fresh	30
Reassigned	Shape within a pass; a new shape for each kind in every pass	Random (0.9); grouped (23)	Fresh	30
Regular	Shape	Random (0.9); graded (1.4, 3.6, 7, 12); grouped (23)	Fixed	90
Regular	Shape	Random (0.9); graded (3.6)	Fixed, 512-token window	30
Mixed-slot	One shape-list word, in the shape slot for about half the kinds and the texture slot for the rest	Random; grouped	Fixed	30
No stable feature	Nothing	Random; grouped; random with a 512-token window	Fixed	45
Colour stable	Colour. Shape varies.	Random; grouped	Fixed	30
Objects	Shape; each object described four times	Random (1.0); same-object runs (3.6, 24); different-object runs (3.6, 24)	Fixed	75
Further controls: no kind label, weak label, paraphrased, frequency matched, added noise, feature swap	As Regular, with one change each	Random	Fixed	90
Total				510

E.2 Models and training

We trained decoder-only transformers of three sizes: 3.4 million parameters (4 layers, 4 heads, width 256), 11.0 million (6 layers, 8 heads, width 384) and 25.6 million (8 layers, 8 heads, width 512). They were trained for 5,000, 8,000 and 10,000 steps with batches of 32, 32 and 16, peak learning rates of 3×10^{-4}, 2×10^{-4} and 1×10^{-4}, AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up, cosine decay and dropout of 0.1. Each training example is a 128-token window, which spans several sentences, so attention crosses sentence boundaries. The tokeniser for the Regular corpora has 351 entries (314 to 339 for the other corpus families), and every artificial word is a single token. A seed sets the corpus, the initial weights and the batch order (the long version lists the models per condition). For the window test, models were trained with 512-token windows and a quarter of the batch size, so that every step saw the same number of tokens. All models have separate input and output embeddings, and the parameter counts include both; the model code ties the two, but the tie was lost on the training hardware (Appendix A). Table C1 lists every setting.

E.3 Measuring the expectation

Eight novel nouns that never occur in training (*snerg*, *twick* and so on) are used in test items. They are in the vocabulary, so each is one token, but their input embeddings were never trained; weight decay shrinks them by less than 10% over training, so they remain distinct random vectors. With the intended tie, the same vectors would have served as the novel nouns' output weights and would have changed in training; we cannot say whether the models would then have tracked these nouns. Novel nouns keep learned knowledge about a noun out of the test, but the test then depends on the model registering that the same untrained token recurs. We therefore also test with nouns from training, whose embeddings are trained (long version). Each item gives the model one sentence in frame 1 describing an exemplar, followed by two continuations that differ in one word:

> Context: *A snerg is a krannish pulek skorli thing.* Repeat: *A snerg is a* ***krannish*** *temil tramel thing.* New: *A snerg is a* ***tova*** *temil tramel thing.*

In this item the probed dimension is shape. The repeat continuation keeps the context's shape, the new one does not, and both give new values for the other two dimensions. The model scores each continuation by the sum of the log probabilities of its tokens given the context, so the difference, Δ, includes the changed word and its effect on the words after it. Δ is the model's preference for repeating the probed dimension, in nats. Trained models generally prefer words they have just seen, so Δ alone mixes the dimension with copying. Our index is the difference between dimensions:

$$\text{index} = \Delta_{\text{shape}} - \Delta_{\text{colour}}$$

Copying that affects shape and colour equally cancels. Copying that differs between dimensions does not, which is one reason every conclusion rests on comparisons between corpora tested in the same way, and why we report Δ for each dimension as well. The shape-over-texture index is defined in the same way. With one exemplar, nothing in the context shows which dimension is fixed, so differences between trained conditions come from training.

Every item is also given with a different novel noun in the continuations (*A yeff is a krannish temil tramel thing*). The **noun-specific index** is the index after the same noun minus the index after a different noun. A kind-tied expectation scores above zero on it; a noun-blind one scores zero.

With 25 draws of features per noun, three probed dimensions and two continuation nouns there are 1,200 one-exemplar items. A further 2,400 items with two or four exemplars test the use of evidence in the context (long version). Untrained models of the same architecture, with the Regular tokeniser, give an index of 0.02 nats (SD 0.07).

E.4 Statistics

Each model gives one index, its mean over items. Models of the three sizes trained with the same seed share a corpus, so the five seeds are the units of replication. We average the three sizes within each seed and compare conditions by paired t tests over the five seed means (4 degrees of freedom); the differences and 95% intervals given in the text are these seed-level estimates. Registered contrasts were tested one-sided, with Holm (1979) correction within each registered family. For fresh-text models the registered test was this seed-level test. For fixed-file models it was a linear mixed model (index $\sim$ condition + size, with a random intercept for seed) or, for some contrasts, a Welch t test over models; the long version gives the seed-level results beside them. Single conditions were compared with zero by one-sample t tests. Results reported with a Holm-corrected p, or called registered, were registered; all others are exploratory. We registered 0.2 nats as the smallest effect of interest and tested equivalence to zero against it with two one-sided tests (Lakens, 2017). This bound is small next to the effects reported here, so the equivalence tests are strict.

F Test-noun representation and changed-word scoring

These analyses were added after the registrations and are exploratory. They rescore the 120 fresh-text models on the one-exemplar items. **Typical tokens**: each novel noun's input embedding is replaced

by the mean input embedding of the 32 training nouns plus $r\,u_j$, where r is the mean distance of the training-noun embeddings from that mean and the u_j are orthonormal random directions (fixed seed), so the eight tokens look like ordinary nouns but stay distinct; output weights are unchanged. **Changed-word Δ**: the difference in log probability of the one word that differs between the continuations, given the identical prefix. With the original embeddings and whole-continuation Δ, the rescoring reproduced the reported scores (144,000 item scores; largest difference 2×10^{-5} nats). Trained-noun values are from the registered trained-noun items (long version in the supplementary material).

Table F1: Index and noun-specific index under four scorings (nats; mean of seed means). Original: unseen tokens with untrained embeddings, as reported; typical: the same tokens with typical embeddings; full: whole-continuation Δ, as reported; word: changed word only.

	Index				Noun-specific index			
Condition	orig./full	typ./full	orig./word	typ./word	orig./full	typ./full	orig./word	typ./word
Regular, random	−1.56	−1.54	−1.26	−1.50	−0.05	−0.31	0.01	−0.20
Regular, graded (3.5)	1.28	−1.41	1.45	−1.25	−0.03	−0.15	0.03	−0.04
Regular, grouped	9.52	1.46	9.65	1.38	0.12	0.00	0.12	0.01
No stable feature, grouped	−3.98	−3.51	−3.74	−3.24	−0.35	−0.26	−0.34	−0.21
Objects, same-object runs	5.87	−1.85	6.06	−1.87	−0.07	0.14	−0.06	0.06
Objects, different-object runs	7.60	−1.25	7.59	−1.33	−0.01	0.19	−0.01	0.11
Reassigned, random	0.21	0.18	0.32	0.27	0.02	−0.10	0.03	−0.01
Reassigned, grouped	9.80	3.87	9.94	3.98	0.37	0.70	0.37	0.74

Table F2: Preference for repeating the context's shape and colour after one exemplar, same noun, for the three kinds of test noun (nats; whole-continuation Δ).

	Δ shape			Δ colour		
Condition	Unseen	Typical	Trained	Unseen	Typical	Trained
Regular, random	−1.55	−1.52	0.03	0.01	0.02	0.05
Regular, graded (3.5)	1.18	−1.52	0.06	−0.10	−0.12	0.19
Regular, grouped	6.88	−0.72	0.06	−2.64	−2.18	−1.94
No stable feature, grouped	−3.88	−3.51	−3.58	0.10	0.00	0.17
Objects, same-object runs	6.87	−1.17	−0.09	0.99	0.68	0.46
Objects, different-object runs	8.05	−1.14	0.04	0.46	0.11	0.01
Reassigned, random	0.22	0.24	0.24	0.02	0.06	0.10
Reassigned, grouped	6.81	1.17	10.01	−2.99	−2.70	−1.65

Table F3: Noun-familiarity curve: unseen tokens' input embeddings blended from their untrained values (0) to typical embeddings (1). Index and Δ shape, same noun (nats; mean of seed means); slope per unit blend with 95% interval over five seeds.

Condition	Measure	0	0.25	0.5	0.75	1	Slope
Regular, random	Index	−1.56	−1.83	−1.93	−1.79	−1.54	0.03 [−0.22, 0.28]
Regular, random	Δ shape	−1.55	−1.83	−1.92	−1.78	−1.52	0.05 [−0.26, 0.35]
Regular, grouped	Index	9.52	6.70	4.32	2.10	1.46	−8.28 [−9.87, −6.70]
Regular, grouped	Δ shape	6.88	4.27	2.08	−0.08	−0.72	−7.82 [−9.59, −6.04]
Objects, same-object runs	Index	5.87	3.71	1.08	−1.41	−1.85	−8.23 [−10.05, −6.40]
Objects, same-object runs	Δ shape	6.87	4.59	1.90	−0.65	−1.17	−8.52 [−10.22, −6.83]
Objects, different-object runs	Index	7.60	4.53	1.87	−0.82	−1.25	−9.22 [−9.97, −8.46]
Objects, different-object runs	Δ shape	8.05	4.85	2.08	−0.69	−1.14	−9.57 [−10.14, −9.00]
Reassigned, grouped	Index	9.80	8.03	5.34	3.87	3.87	−6.41 [−8.74, −4.08]
Reassigned, grouped	Δ shape	6.81	4.83	2.24	0.97	1.17	−6.06 [−7.28, −4.84]

Mean input-embedding norms: trained nouns 0.55, unseen tokens 0.36, typical tokens 0.55.

G Terms, hypotheses and registrations

Terms. The registrations and the detailed appendices that follow use the labels of the registration documents. Table G1 gives the correspondence with the terms of the main text.

Table G1: Main-text terms and registration terms.

Main text	Registrations and Appendices H, I and J
Experiment 1	Amendment 9 (the familiarity study)
Experiment 2	Amendment 10, Part A
Experiment 3	Amendment 11 (Parts A, B and C)
Single-history corpora (changing-only, fixed-only)	pure_changing, pure_fixed
Mixed corpora (Experiment 3)	Regular grouped, mixed design
Row analysis (Section 4.1)	Amendment 10, Part C
Pretrained models (Section 4.10)	Amendment 10, Part B
Stable-shape corpora	Regular (grouped) corpora
Shifting-shape corpora	Reassigned (grouped) corpora
Spaced, bursts	singles, runs
Changing shape	uninformative (Experiment 1), rotating (Experiments 2 and 3)
No shape	shape-absent
Copy score	carry
Typical embedding	typical row
Random embedding of untrained scale	atypical_init

Hypotheses. Table G2 lists every registered hypothesis of Experiments 1, 2 and 3, the row analysis and the pretrained test, the claim of the main text it bears on, and its outcome under the registered decision rules (Holm correction within each experiment; direction in at least two of three model sizes; for the pretrained test, Holm within each model and support in at least four of five models). Equivalence bounds were set in each registration: ± 1 nat in Amendment 9 (about one seventh of the 7.6-nat fall from unseen to typical embeddings in the earlier experiments); in Amendment 10, ± 0.3 nats for H40, H41 and H44 (the smallest history effect seen in Amendment 9), ± 1 nat for the check K1, and the larger of 0.3 nats and one fifth of the pilot effect for H48; in Amendment 11, ± 0.3 nats for H52, H54 and H59, kept from Amendment 10. The hypotheses of the earlier experiments are in Appendix A.

Table G2: Registered hypotheses and outcomes. Full statistics: Tables H1 (Experiment 1), I1 (Experiment 2), I10 (row analysis), I11 (pretrained models) and J1 (Experiment 3).

Label	Prediction (main-text terms)	Outcome
Experiment 1 (Amendment 9)		
M1	Fixed-shape nouns store their shape (knowledge $>$ 0.2 nats at 1,000 exposures, rising)	Passed in all three corpora
H30	Copying falls faster with exposure for fixed-shape than no-shape nouns	Not supported (wrong direction)
H31	No-shape nouns (1,000 and 10,000) differ from unseen nouns with a typical embedding; equivalence ± 1	Different (0.33) and equivalent
H32	Fixed-shape nouns' preference for their own shape, measured in the copy items, rises with exposure	Supported
H33	Ablating candidate heads removes more than half of unseen nouns' copying, more than random heads	Not supported
H34	Ablating candidate heads lowers trained kinds' copying in shifting-shape corpora	Supported; random heads cut it nearly as much
H35	Candidate heads' attention falls faster for fixed-shape nouns	Not tested (required H33)
R1	Unseen nouns copy more in grouped than random order	Replicated
Experiment 2 (Amendment 10, Part A); stable-shape corpora		
M2	Unseen nouns copy more than 1 nat	Passed
K1	Fixed-shape nouns' knowledge equal (± 1) in bursts and spaced	Not equivalent: H41 not interpreted
H40	No-shape: bursts differ from spaced; equivalence ± 0.3	Bursts higher
H41	Fixed shape: bursts differ from spaced; equivalence ± 0.3	Bursts higher (not interpreted)
H42	Changing shape: bursts copy more than spaced	Supported
H43	Spaced: fixed shape copies less than no shape	Supported; confounded by expected word class (Section 4)
H44	Spaced no-shape nouns differ from unseen nouns with a typical embedding; equivalence ± 0.3	Higher; not equivalent
Experiment 3 (Amendment 11); stable-shape corpora; Holm over H50–H53, H55, H56		
M3	Unseen nouns copy more than 1 nat	Passed
H50	At 1,000: changing-shape bursts copy more than fixed-shape bursts (primary)	Supported (3.68)
H51	Changing shape: bursts copy more than spaced	Supported (4.14)
H52	Spaced: fixed differs from changing shape; equivalence ± 0.3	Fixed lower (-0.25); not equivalent
H53	Fixed-shape spaced nouns prefer their own shape to the context's	Supported (5.84)
H54	Noun-specific copying of changing-shape bursts within ± 0.3	Within ± 0.3 (0.08)
H55	Changing-shape bursts copy more than row geometry predicts	Supported (3.67)
H56	The queried noun's history matters more than the context noun's	Supported (1.25)
M3B, H57	85.6M models: unseen nouns copy; changing-shape bursts copy more than fixed-shape bursts	Pending
M3C	Single-history corpora: unseen nouns copy	Passed
H58	Changing-shape bursts (changing-only) copy more than fixed-shape bursts (fixed-only)	Supported (5.51)
H59	Single-history minus mixed corpora, same history; equivalence ± 0.3	Changing bursts inconclusive; fixed bursts lower (-1.00)
Row analysis (Amendment 10, Part C); Experiment 1 models		
H49	Scaling unseen nouns' rows to the trained kinds' norm lowers copying	Supported (driven by the 3.4M models)
Pretrained models (Amendment 10, Part B)		
H45	A random embedding of untrained scale raises copying over the own embedding	Not supported (2 of 5 models)
H46	The same embedding raises copying over a typical embedding	Not supported (2 of 5 models)
H47	Colour-typical nouns copy less than colour-variable nouns	Not supported (0 of 5)
H48	The H45 increase differs between colour-typical and colour-variable nouns	Inconclusive

H The familiarity study (Amendment 9)

Registration. The design, hypotheses and analysis were registered before any corpus was built (Amendment 9). An addendum, written the same day before any corpus was built, fixed details the registration left open (36 context draws per noun; uniform frame proportions; exposures placed at evenly spaced read positions from 5% to 95% of training, after warm-up for every size; random in-window position; the 40 nonce nouns; how the pilot fixes the number of nouns per cell; how a hypothesis counts as supported; the gate for H35; and that heads without prefix matching are described as copying heads). A pilot (one seed, 3.4M models, never used in a test) set three nouns per cell; with the registered rule the expected power for a 1-nat difference was 0.57, below the planned 0.8, which we report as a shortfall. Corpus checksums were recorded before training and matched on the training machine; a checker verified every exposure, window and frame from the tokens. All models were trained on one machine.

Design. In every model, 40 nonce nouns fill 13 cells (three per cell, four unseen): fixed shape, uninformative shape (a new shape in every sentence, drawn from the corpus's shape distribution) and shape-absent (a sixth frame, *The noun is colour and texture*, also used by trained kinds in one sentence in fifteen), each at 10, 100, 1,000 and 10,000 exposures, plus unseen. Nouns rotate over cells across ten seeds. Each familiarity sentence lies wholly inside one 128-token window, at most one per window, between runs in grouped corpora. Base corpora: Regular grouped, Regular random, Reassigned grouped; sizes 3.4M, 10.9M and 25.6M; ten seeds; 90 models with untied embeddings. Measures are read from the full log-softmax over the ten shape words at the position that predicts the shape in *A noun is a*: carry $= L(s) - \bar{L}$ after a context sentence with shape s; stored preference $= L(s^*) - \bar{L}$; knowledge $= L(s^*)$ minus the mean over the other nine shapes after an unrelated sentence, where $\bar{L}$ is the mean over shapes other than s and s^*.

Table H1: Registered tests of the familiarity study (Regular grouped models unless stated; seed-level estimates over ten seeds, sizes averaged within seed; nats, proportions for H33a, attention probability for H35). Holm p is adjusted over H30 to H34; H35 was not entered because H33(a) failed. Sizes: number of the three model sizes with the registered direction.

	Test	Estimate	95% CI	Holm p	Sizes	Verdict
M1	Knowledge at 1,000 exposures, fixed shape (each corpus)	9.53; 8.61; 8.26				Passed
H30	Carry falls faster with exposure for fixed-shape than shape-absent nouns (slope difference)	0.16	[0.01, 0.30]	1.000	0	Not supported
H31	Shape-absent (1k and 10k) minus unseen with typical embedding (two-sided; TOST ± 1)	0.33	[0.13, 0.52]	.013	3	Different, and equivalent within 1 nat
H32	Stored preference rises with exposure, fixed shape (slope)	2.97	[2.79, 3.16]	$<.001$	3	Supported
H33a	Ablating candidates removes more than half of unseen carry (proportion)	0.22	[0.07, 0.36]	1.000	2	Not supported
H33b	Random heads minus candidates, unseen carry after ablation	0.64	[−0.43, 1.71]	1.000	2	Not supported
H34	Reassigned: trained-kind carry intact minus candidates ablated	7.41	[6.02, 8.80]	$<.001$	3	Supported
H35	Attention slope difference, fixed shape minus shape-absent	0.002	[0.000, 0.005]	–	2	Not tested (gate)
R1	Replication: grouped minus random Regular, unseen carry	4.68	[3.71, 5.65]	$<.001^a$	3	Replicated

[a] Outside the Holm family (replication); unadjusted one-sided p. M1 values: Regular grouped; Regular random; Reassigned grouped (all with slopes above zero, $p < .001$). H31 TOST $p < .001$.

Table H2: Carry (nats) by noun type and exposure, mean of seed values (sizes averaged). Unseen: never in training; typical: the unseen nouns with a typical embedding; trained: the 32 base kinds.

Corpus	Unseen	Typical	Trained	Familiar nouns	10	100	1k	10k
Regular grouped	4.53	−0.33	−0.02	Fixed shape	3.70	0.30	−0.30	−0.37
				Uninformative	3.57	0.16	0.00	−0.01
				Shape-absent	4.47	0.86	0.05	−0.05
Reassigned grouped	6.85	1.57	9.98	Fixed shape	6.40	2.13	0.70	0.60
				Uninformative	6.34	2.53	0.30	0.15
				Shape-absent	6.97	4.89	2.04	1.86
Regular random	−0.15	−0.72	−0.10	Fixed shape	−0.13	−0.13	−0.06	−0.05
				Uninformative	−0.10	−0.03	−0.02	−0.01
				Shape-absent	−0.19	−0.09	−0.07	−0.05

Table H3: Knowledge and stored preference (nats) in Regular grouped models, mean of seed values.

Type	Measure	10	100	1k	10k
Fixed shape	Knowledge	1.59	8.02	9.53	10.61
	Stored	1.41	7.74	9.56	10.71
Uninformative	Knowledge	0.04	−0.03	0.00	0.03
	Stored	0.01	−0.03	−0.01	0.03
Shape-absent	Knowledge	−0.14	−0.17	0.12	0.08
	Stored	0.11	−0.16	0.10	0.06

Table H4: Head analysis by corpus and size (means over ten seeds). Candidates were the k heads with the highest product of prefix-matching and copying scores; in grouped models they were in layer 0 in all but one model (a 3.4M Regular grouped model chose one layer-1 head), and in random-order models a few were in later layers. In the 3.4M and 10.9M models layer 0 has exactly k non-candidate heads, so the random draws repeat one set. Prefix: highest prefix-matching score of any head (uniform attention gives about 0.027). Carry of unseen nouns and of trained kinds intact, with the candidates mean-ablated, and with random heads ablated (mean of 200 or 20 draws). DLA: change in unseen carry predicted by the candidates' direct paths.

				Unseen carry			Trained carry			
Corpus	Size	k	Prefix	Intact	Cand.	Random	Intact	Cand.	Random	DLA
Regular grouped	3.4M	2	0.028	8.10	−0.42	5.26	−0.25	−0.85	−0.71	−0.04
	10.9M	4	0.037	2.98	5.80	1.95	0.08	0.17	−0.40	−0.01
	25.6M	6	0.037	2.50	0.73	0.81	0.12	−0.60	−0.18	−0.01
Reassigned grouped	3.4M	2	0.029	6.48	4.22	3.27	9.59	2.83	3.37	−0.12
	10.9M	4	0.032	7.21	6.56	4.82	10.10	5.02	2.45	−0.03
	25.6M	6	0.031	6.88	0.26	3.26	10.26	−0.11	3.39	−0.01
Regular random	3.4M	2	0.028	−0.24	−0.09	−0.34	−0.08	0.00	−0.11	−0.01
	10.9M	4	0.030	0.04	0.25	0.03	−0.06	0.04	−0.11	−0.01
	25.6M	6	0.030	−0.25	0.00	−0.37	−0.18	−0.10	−0.27	−0.01

Table H5: Attention of the candidate heads from the position that predicts the shape to the context's shape word (mean over candidates; mean of seed values; exploratory).

Corpus	Type	Unseen	10	100	1k	10k
Regular grouped	Fixed shape	0.155	0.144	0.082	0.071	0.071
	Uninformative		0.147	0.088	0.076	0.074
	Shape-absent		0.154	0.097	0.077	0.075
Reassigned grouped	Fixed shape	0.219	0.217	0.108	0.055	0.059
	Uninformative		0.217	0.157	0.071	0.070
	Shape-absent		0.219	0.170	0.115	0.139
Regular random	Fixed shape	0.089	0.088	0.076	0.067	0.067
	Uninformative		0.087	0.082	0.075	0.073
	Shape-absent		0.089	0.084	0.075	0.073

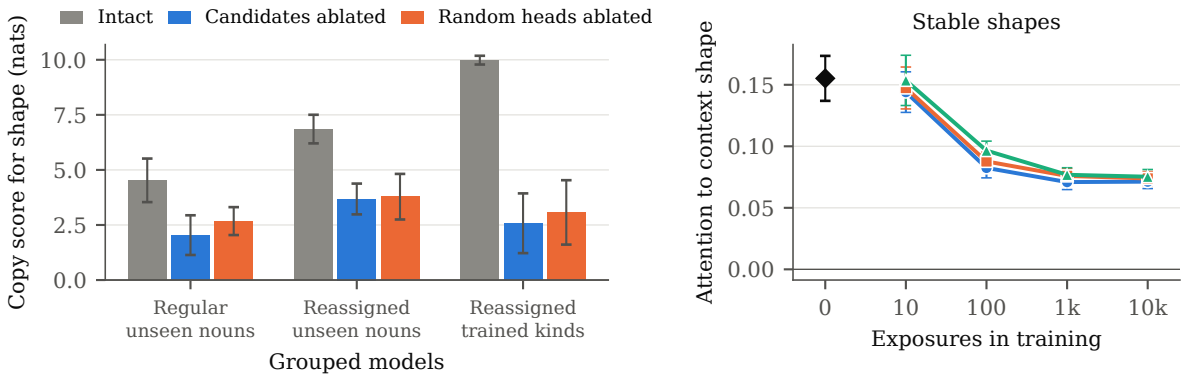


Figure H1: Experiment 1. Left: copy score intact, with the candidate heads mean-ablated, and with random heads ablated (nats; mean over ten seeds and three sizes, 95% intervals over seeds). Right: the candidates' attention from the position that predicts the shape to the context's shape word, stable-shape models (exploratory).

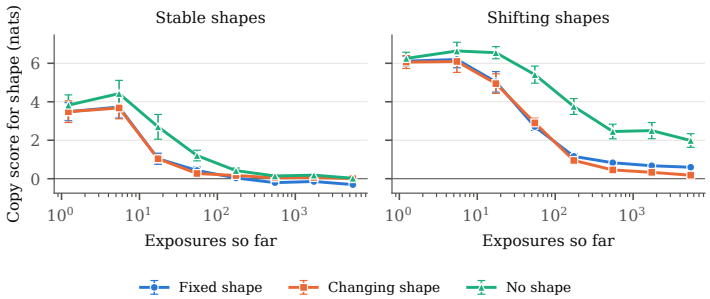


Figure H2: Experiment 1. Copy score against the exposures each noun had received by each saved checkpoint (exploratory; nats; mean over ten seeds of size-averaged values, 95% intervals). Exposure is confounded with training time.

I Recurrence, matched frames, embedding rows and pretrained models (Amendment 10)

Registration. Amendment 10 has three parts: Part A is Experiment 2, Part B the pretrained test and Part C the row analysis. It was registered after the Amendment 9 results, after its corpora were built and checksummed, and before any of them was used, any of its models was trained, or any pretrained model was run on its items. It answered two confounds raised by a blind review of the Amendment 9 results (by an AI assistant given only the write-up of Experiment 1): familiarity nouns never recurred within a window, and shape-absent nouns never appeared in a frame with a shape slot after *is a*. The noun lists for Part B were fixed, with checksums, before the pilot. A pilot on Pythia-410M (never in a test) gave a mean increase of 0.13 nats from the original to the atypical_init row, so the smallest effect of interest for H48 was 0.30 nats by the registered rule. One deviation: some Pythia tokenizers define no padding token, so the scoring script falls back to token id 0; padded positions are masked and only the last real token is read, and batched and single-item scores agreed exactly. No score had been produced by the affected models.

Part A design. Everything not stated is as in Amendment 9 (Appendix H). Histories: *fixed*, frame 1 (*A noun is a shape colour texture thing*) with the noun's own shape; *absent*, frame 7 (*A noun is a colour texture thing*), which names no shape; *rotating*, frame 1 with a shape drawn from the corpus's shape distribution, a new one for every sentence when presented singly and one per run, changing between runs, when presented in runs. Presentation: *single*, one sentence per unit; *run*, four consecutive sentences about the noun in one window. Exposures (100 or 1,000) count sentences, so runs and singles are matched in sentences, not in units or spacing. Trained kinds use frame 7 in one sentence in fifteen, so the frame is not new. Three nouns per cell (12 cells) and four unseen nouns; Regular grouped (confirmatory) and Reassigned grouped (exploratory) base corpora; three sizes and ten seeds (60 models). A checker re-read every corpus and verified exposures, units, frames, shapes within runs and window placement; all 40

corpora matched checksums recorded before training.

Table I1: Registered tests of Part A (Regular grouped models; 100 and 1,000 exposures pooled; seed-level estimates over ten seeds, sizes averaged within seed; nats). Holm p is adjusted over H40 to H44. Sizes: number of the three sizes with the estimate's direction. Readings follow the registered table: "above" means the Holm threshold was met and the direction held at two of three sizes; TOST p for H40, H41 and H44 was above .99, so none was equivalent.

	Test	Estimate	95% CI	Holm p	Sizes	Reading
M2	Unseen carry minus 1 nat (manipulation check)	5.35	[4.59, 6.12]	$<.001^a$		Passed
K1	Knowledge, fixed runs minus fixed singles (equivalence ± 1)	−1.09	[−1.45, −0.73]	$<.001^a$		Not equivalent (H41 not interpreted)
H40	Absent runs minus absent singles (two-sided; TOST ± 0.3)	0.80	[0.47, 1.13]	$<.001$	3	Above
H41	Fixed runs minus fixed singles (two-sided; TOST ± 0.3)	1.69	[1.29, 2.08]	$<.001$	3	Above (not interpreted)
H42	Rotating runs minus rotating singles (one-sided)	3.39	[2.69, 4.09]	$<.001$	3	Supported
H43	Fixed singles minus absent singles (one-sided)	−1.02	[−1.27, −0.76]	$<.001$	3	Supported
H44	Absent singles minus unseen with typical row (two-sided; TOST ± 0.3)	1.42	[1.12, 1.73]	$<.001$	3	Above

[a] Outside the Holm family; unadjusted p. The M2 estimate is unseen carry minus 1 nat, so unseen carry was 6.35 nats.

Table I2: Carry (nats) by cell, mean of seed values (sizes averaged) with 95% intervals over ten seeds. Unseen nouns have no exposures; the typical row replaces an unseen noun's input row with one typical of trained nouns.

Corpora	Noun	100 exposures	1,000 exposures
Stable shapes	Fixed shape, spaced	1.52 [1.18, 1.86]	0.52 [0.36, 0.69]
	Fixed shape, bursts	4.43 [3.49, 5.36]	0.99 [0.68, 1.30]
	No shape, spaced	3.09 [2.56, 3.63]	0.98 [0.76, 1.20]
	No shape, bursts	4.65 [3.94, 5.36]	1.03 [0.59, 1.47]
	Changing shape, spaced	1.59 [1.30, 1.88]	0.50 [0.38, 0.62]
	Changing shape, bursts	5.02 [4.18, 5.86]	3.85 [2.93, 4.77]
	Unseen	6.35 [5.59, 7.12]	
	Unseen, typical embedding	0.61 [0.28, 0.95]	
Shifting shapes	Fixed shape, spaced	2.23 [1.92, 2.55]	0.44 [0.37, 0.51]
	Fixed shape, bursts	5.35 [4.88, 5.82]	3.57 [3.02, 4.11]
	No shape, spaced	1.83 [1.49, 2.17]	0.63 [0.53, 0.72]
	No shape, bursts	3.79 [3.38, 4.20]	2.09 [1.82, 2.35]
	Changing shape, spaced	2.37 [2.04, 2.71]	0.51 [0.38, 0.64]
	Changing shape, bursts	5.40 [4.95, 5.86]	6.82 [6.48, 7.17]
	Unseen	5.02 [4.61, 5.42]	
	Unseen, typical embedding	0.43 [0.32, 0.55]	

Table I3: Knowledge of the noun's own shape and expectation of a shape rather than a colour at the query position (slot: log of the summed probability of the ten shapes minus that of the ten colours), Regular grouped models, mean of seed values (exploratory). Unseen nouns: knowledge −0.04, slot 1.49.

	Knowledge		Slot	
Noun	100	1,000	100	1,000
Fixed shape, singles	5.37	7.80	4.03	6.16
Fixed shape, runs	3.02	7.98	3.28	6.24
Shape-absent, singles	0.00	−0.09	−2.98	−4.93
Shape-absent, runs	0.05	−0.25	−1.12	−4.43
Rotating shape, singles	−0.08	0.01	5.04	5.30
Rotating shape, runs	0.14	0.06	3.74	5.24

Reanalysis of the Experiment 2 models (exploratory, added after the results). For every familiar noun in the 60 final models we recorded its input row's cosine with its initial row (rebuilt from the seed;

Table I4: The Part A contrasts in Reassigned grouped models (exploratory; seed-level, unadjusted two-sided p).

	Contrast	Estimate	95% CI	p
H40	Absent runs minus absent singles	1.71	[1.36, 2.06]	<.001
H41	Fixed runs minus fixed singles	3.12	[2.71, 3.53]	<.001
H42	Rotating runs minus rotating singles	4.67	[4.28, 5.06]	<.001
H43	Fixed singles minus absent singles	0.11	[−0.07, 0.29]	.196

Table I5: Matched contrasts in Part A (exploratory; seed-level, nats, 95% intervals). Rotating minus fixed runs: same frame, presentation and number of occasions; the two differ only in whether the shape changes between runs (and so in whether a shape is stored). Fixed minus rotating singles: same frame, presentation and expectation of a shape at the test position (Table I3); they differ only in whether the noun has one stored shape.

Base	Contrast	Pooled	100	1,000
Regular grouped	Rotating runs − fixed runs	1.73 [1.25, 2.20]	0.59 [0.23, 0.95]	2.86 [1.99, 3.74]
	Fixed singles − rotating singles	−0.02 [−0.16, 0.11]	−0.07 [−0.29, 0.15]	0.02 [−0.04, 0.08]
	Fixed runs − fixed singles	1.69 [1.29, 2.08]	2.91 [2.21, 3.60]	0.47 [0.27, 0.66]
Reassigned grouped	Rotating runs − fixed runs	1.65 [1.35, 1.96]	0.05 [−0.05, 0.16]	3.26 [2.64, 3.87]
	Fixed singles − rotating singles	−0.10 [−0.23, 0.02]	−0.14 [−0.34, 0.06]	−0.07 [−0.20, 0.05]
	Fixed runs − fixed singles	3.12 [2.71, 3.53]	3.12 [2.61, 3.62]	3.13 [2.59, 3.66]

Knowledge, fixed runs minus fixed singles (Regular grouped): −2.35 [−2.82, −1.88] at 100 exposures, 0.18 [−0.18, 0.54] at 1,000.

Table I6: Noun-specific carry (exploratory): carry when the context sentence is about the query noun minus carry when it is about another noun of the same cell, with the same shape (nats; seed-level means). Near zero means the model copies the context's shape whichever noun the context was about.

		Regular grouped		Reassigned grouped	
Experiment	Noun	100	1,000	100	1,000
2	Unseen	0.05		0.48	
	Fixed, singles	0.01	−0.02	0.22	0.06
	Fixed, runs	0.06	−0.05	0.58	0.65
	Shape-absent, singles	0.07	0.05	0.25	0.05
	Shape-absent, runs	0.01	0.00	0.46	0.70
	Rotating, singles	0.04	0.01	0.35	0.05
	Rotating, runs	0.05	0.05	0.58	1.45
1	Unseen	0.14		0.45	
	Fixed shape	−0.05	−0.02	0.21	0.01
	Uninformative	0.00	0.00	0.51	0.04
	Shape-absent	0.05	0.00	0.45	0.22

Table I7: Part A by model size (Regular grouped; seed means, 100 and 1,000 exposures pooled; nats).

Size	Unseen	Typical row	H40	H41	H42	H43	H44	Rot. − fixed runs
3.4M	7.67	−0.30	0.65	2.41	5.05	−0.84	1.67	2.94
10.9M	5.74	0.64	1.23	1.46	2.82	−0.78	1.25	1.24
25.6M	5.65	1.51	0.53	1.20	2.29	−1.43	1.36	1.00

the rebuild reproduced the unseen nouns' rows, which receive no gradient, in every model), its distance from the trained kinds' mean row and its norm; the copy score with its own row, with a typical row and with the row of a noun of another history (rows swapped between changing- and fixed-shape bursts and between changing-shape bursts and spaced nouns); the copy score with one sentence about a trained kind between context and query (gap), with the context in the frame *Look at the noun. It is shape and colour.* (other frame), and for the context's colour after *A noun is a shape* (colour); and first-layer attention from the query position to the context's shape, averaged over heads. Full results, including shifting-shape corpora and 100 exposures, are in the supplementary material.

Part C (embedding rows in the Amendment 9 models). In the 60 final grouped models of Amendment 9 we replaced the input rows of unseen nouns and scored carry again; the unmodified condition reproduced

Table I8: Experiment 2 reanalysis, stable-shape corpora, 1,000 exposures (unseen nouns: no exposures). Copy scores in nats; cosine of the row with its initial value; first-layer attention from the query to the context's shape. Seed-level means. Swapping rows between changing- and fixed-shape bursts moved the copying with the row (2.88 nats [2.01, 3.76] in each direction). In a noun-level regression of the copy score on row geometry (cosine with the initial row, distance from the trained kinds' mean, norm, size and seed; standard errors clustered by seed), changing-shape bursts scored 3.19 nats [2.46, 3.91] more than geometry predicts at 1,000 exposures and −0.09 [−0.51, 0.33] at 100.

Noun	Own row	Typical row	Gap	Other frame	Colour	Cos. initial	Attention
Unseen	6.35		0.16	4.38	−0.30	1.000	0.186
Fixed, spaced	0.52	0.67	0.05	0.12	−0.10	0.732	0.089
Fixed, bursts	0.99	0.61	0.13	0.32	−0.12	0.778	0.099
No shape, spaced	0.98	0.63	−0.04	0.71	−0.22	0.793	0.102
No shape, bursts	1.03	0.68	−0.07	0.66	−0.34	0.806	0.097
Changing, spaced	0.50	0.61	0.10	0.20	−0.10	0.794	0.108
Changing, bursts	3.85	0.66	0.37	1.53	−0.12	0.849	0.168

Table I9: Crossed histories (exploratory), stable-shape corpora, 1,000 exposures (unseen nouns: none): copy score when the context sentence is about a noun of one history (rows) and the query names a different noun of another (columns); 24 items per pair and model, seed-level means. Averaged over familiar histories, a queried noun with changing-shape bursts copied 1.35 nats [0.97, 1.74] more than one with fixed-shape bursts, and a context noun with changing-shape bursts raised copying by 0.39 [0.19, 0.58]. Full results are in the supplementary material.

Context \ query	Unseen	Fixed spaced	Fixed bursts	No shape spaced	No shape bursts	Changing spaced	Changing bursts
Unseen	6.34	0.60	0.73	1.16	1.01	0.42	2.61
Fixed, spaced	3.67	0.62	0.75	0.94	0.93	0.44	1.73
Fixed, bursts	5.14	0.68	1.01	1.07	1.27	0.53	2.45
No shape, spaced	4.10	0.53	0.74	1.12	1.13	0.39	1.67
No shape, bursts	4.30	0.62	0.91	0.89	1.01	0.43	2.07
Changing, spaced	4.03	0.59	0.91	1.26	1.12	0.50	1.86
Changing, bursts	6.49	0.85	1.15	1.49	1.39	0.66	3.80

the Amendment 9 scores. Conditions: own row; own direction scaled to the mean norm of the trained kinds' rows (ownnorm); a random direction at that norm (randnorm); the trained kinds' mean row. Shape-absent nouns at 10,000 exposures were also scored with their own direction scaled to the unseen nouns' mean norm (shrunk).

Table I10: Part C: carry (nats) by input row, mean over seeds (sizes averaged). H49 (registered, Regular grouped): ownnorm minus own −0.52 [−0.85, −0.20], one-sided $p = .003$, direction at 2 of 3 sizes, supported. In Reassigned models the same contrast went the other way (0.27 [0.12, 0.41]; exploratory). Randnorm minus ownnorm: 0.09 (Regular), 0.03 (Reassigned).

	Unseen nouns				Shape-absent, 10k	
Corpus	Own	Ownnorm	Randnorm	Trained mean	Own	Shrunk
Regular grouped	4.53	4.00	4.09	−0.44	−0.05	0.53
Reassigned grouped	6.85	7.12	7.15	1.05	1.86	4.16

Part B (pretrained models). 44 nouns in 22 pairs, matched on Pile frequency and count/mass type, were coded for colour typicality by two independent AI passes with a fixed prompt before any model was run (exact agreement 94%, weighted $\kappa = 0.96$); nouns that are also colour terms or that both passes flagged as polysemous were excluded. Colour-typical nouns are mostly natural kinds, so kind was a covariate. Each noun was read in three neutral frames with each of the eleven basic colour terms that are one token; carry is the mean over frames and ordered colour pairs of $\log p(c_1 \mid c_1) - \log p(c_1 \mid c_2)$, with colour probabilities renormalised over the colour set. Only the noun's input row was changed.

Table I11: Part B registered tests by primary model (nouns or pairs as units; Holm within model over H45 to H48). H45: atypical_init minus original; H46: atypical_init minus typical row; H47: colour-typical minus colour-variable nouns, original rows; H48: difference between groups in the H45 increase (TOST against ± 0.30). A hypothesis needed Holm in at least four of the five models. Outcome: H45 and H46 met Holm in two of five models, H47 in none, and H48 was inconclusive (equivalent in one model).

	H45		H46		H47		H48	
Model	Est.	Holm	Est.	Holm	Est.	Holm	Est.	TOST p
1B	0.20	met	0.21	met	0.00	–	0.07	.097
1.4B	0.34	met	−0.22	–	−0.01	–	−0.08	.021
2.8B	−0.38	–	0.08	–	−0.16	–	0.11	.222
6.9B	0.07	–	0.36	met	−0.09	–	0.10	.127
12B	−0.35	–	−0.21	–	0.11	–	0.08	.066

Mixed model over the five primary models (carry $\sim$ condition $\times$ group + model + kind; random intercepts for noun and pair, random condition slopes by noun): atypical_init 0.00 (p = .956), typical row −0.04 (p = .559), colour-variable group 0.06 (p = .527).

Table I12: Part B: carry of the context's colour (nats; mean over nouns) by model and input row, and with original rows by group. Atypical: the atypical_init direction at the noun's own norm (exploratory). 70M and 160M are descriptive. Prior entropy: entropy of the colour distribution after *The noun was* with no context (nats; uniform over eleven colours is 2.40).

Model	Original	Atypical_init	Atypical	Typical row	Colour-typical	Colour-variable	Prior entropy
70M	1.59	1.40	1.41	1.16	1.61	1.56	2.09
160M	0.93	1.95	1.95	1.03	0.87	1.00	2.01
1B	1.89	2.09	2.09	1.88	1.89	1.89	1.95
1.4B	2.10	2.44	2.44	2.66	2.10	2.10	1.97
2.8B	2.70	2.31	2.31	2.23	2.61	2.78	1.92
6.9B	2.67	2.75	2.75	2.39	2.63	2.72	1.94
12B	2.68	2.33	2.33	2.54	2.73	2.62	1.91

Table I13: Part B checks (exploratory). Stored colour: with no context, log probability of a colour-typical noun's coded colour minus the mean over the other colours after *The noun was* (nats; mean over 22 nouns, and the share of nouns above zero). TOST: equivalence of the H45 and H46 differences within ± 0.30 nats over nouns. Norm only: atypical minus atypical_init (same direction, different norm).

Model	Stored colour	Share > 0	H45 TOST p	H46 TOST p	Norm only
70M	0.72	0.73	.059	.129	0.014
160M	0.90	0.77	1.000	1.000	0.000
1B	1.28	0.82	.125	.091	0.001
1.4B	1.30	0.91	.736	.172	0.000
2.8B	1.55	0.86	.776	.001	0.000
6.9B	1.66	0.95	.012	.801	−0.002
12B	1.73	0.86	.740	.023	−0.001

Table I14: Part B, Pythia-1.4B deduplicated across training (exploratory): carry (nats; mean over nouns) by training step and input row.

Row	1,000	2,000	4,000	8,000	16,000	32,000	64,000	143,000
Original	0.73	1.09	0.72	1.97	1.46	1.67	2.39	2.10
Atypical_init	0.81	1.45	1.10	2.08	1.82	2.11	2.56	2.44
Typical row	0.72	1.06	0.60	1.64	1.17	1.70	2.70	2.66

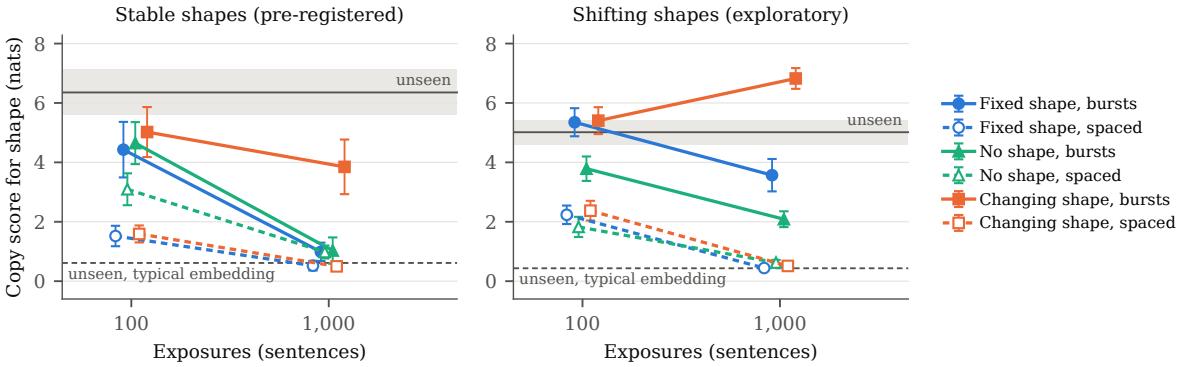


Figure I3: Experiment 2: copy score by history (colour) and presentation (solid: bursts; dashed: spaced) at 100 and 1,000 exposures, in stable-shape (left, pre-registered tests) and shifting-shape (right, exploratory) corpora (means over ten seeds, sizes averaged; 95% intervals). Solid line and band: unseen nouns; dashed line: unseen nouns with a typical embedding.

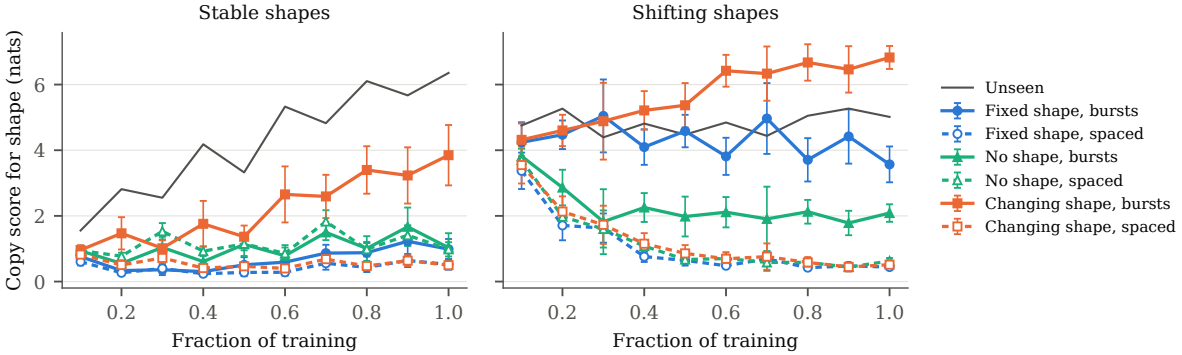


Figure I4: Experiment 2. Copy score over training for nouns with 1,000 exposures, by history and presentation, and for unseen nouns (exploratory; saved weights every 1,000 steps; means over ten seeds, sizes averaged; 95% intervals). A noun with 1,000 exposures has had about $1{,}000 \times (t - 0.05)/0.9$ sentences by fraction t of training, so exposure and training time rise together. Small zig-zags arise because the three sizes save weights at different fractions of training.

J Replication, single-history corpora and a larger model (Amendment 11)

Registration. Amendment 11 is Experiment 3. It was registered (osf.io/qj9hb) after all results of Experiment 2 and its exploratory analyses, after its corpora were built and checksummed, and before any of its models was trained, to test the exploratory findings of Experiment 2 on new models. Part A repeated the stable-shape (Regular grouped) design of Experiment 2 with ten new seeds at 3.4M, 10.9M and 25.6M parameters (30 models). Part B trained ten models of 85.6M parameters (12 layers, 12 heads, width 768) on the same corpora as the 25.6M models. Part C trained five seeds at each of the three smaller sizes on corpora in which all 36 familiar nouns had changing shapes (pure_changing) or all had fixed shapes (pure_fixed), nine nouns per cell, with the same exposures and share of text as the mixed design (30 models). The assignment of noun forms to cells follows the Experiment 2 rule by seed position; model weights, reading order and placement are new. All 40 corpora passed a checker and matched checksums recorded before training. Registered rules for failed models (non-finite loss, or a final loss more than 0.5 nats above the median of its size and design) and for complete seeds were applied; no model failed. Part A used Holm correction over H50, H51, H52 (difference), H53, H55 and H56, with direction required at two of three sizes; equivalence tests used two one-sided tests against ± 0.3 nats. H55 is a seed-level test: within each model, the copy scores of the 18 familiar nouns at 1,000 exposures were regressed on the row's cosine with its initial value, its relative distance from the trained kinds' mean row, its norm and a changing-burst indicator, and the indicator's coefficient was averaged over sizes within seed. There were

no deviations.

Table J1: Registered tests of Experiment 3 (seed-level estimates, nats; sizes averaged within seed; t-based 95% intervals). Holm p: adjusted over the six Part A tests marked ^{h}; other p values are unadjusted. Sizes: number of the three sizes with the estimate's direction.

	Test	Estimate	95% CI	p	Sizes	Reading
M3	Unseen nouns' copy score minus 1 nat (manipulation check; one-sided)	5.88	[5.31, 6.45]	$<.001$		Passed
H50^h	At 1,000: changing-shape bursts minus fixed-shape bursts (one-sided; primary)	3.68	[2.99, 4.37]	$<.001$	3	Supported
H51^h	Changing shape: bursts minus spaced, pooled (one-sided)	4.14	[3.60, 4.69]	$<.001$	3	Supported
H52^h	Spaced: fixed minus changing shape, pooled (two-sided; TOST $\pm$0.3)	−0.25	[−0.35, −0.15]	$<.001$	3	Different (fixed lower); not equivalent
H53^h	Fixed-shape spaced nouns: $L(s^*) - L(s)$, pooled (one-sided)	5.84	[5.57, 6.11]	$<.001$	3	Supported
H54	At 1,000: noun-specific copy score of changing-shape bursts (TOST $\pm$0.3)	0.08	[0.02, 0.14]	.015		Within $\pm$0.3
H55^h	At 1,000: changing-shape bursts minus row-geometry prediction (one-sided)	3.67	[3.03, 4.32]	$<.001$	3	Supported
H56^h	At 1,000: queried-noun minus context-noun effect of changing vs fixed bursts (one-sided)	1.25	[0.94, 1.56]	$<.001$	3	Supported
M3B	85.6M models: unseen copy score minus 1 nat (one-sided)	–	–	–		Pending
H57	85.6M models, at 1,000: changing-shape bursts minus fixed-shape bursts (one-sided)	–	–	–		Pending
M3C	Changing-only corpora: unseen copy score minus 1 nat	5.16	[3.25, 7.08]	$<.001$		Passed
M3C	Fixed-only corpora: unseen copy score minus 1 nat	3.87	[2.84, 4.90]	$<.001$		Passed
H58	At 1,000: changing-shape bursts (changing-only) minus fixed-shape bursts (fixed-only) (one-sided)	5.51	[4.24, 6.78]	$<.001$	3	Supported
H59	At 1,000: changing-shape bursts, changing-only minus mixed corpora (two-sided; TOST $\pm$0.3)	0.24	[−1.44, 1.92]	.712		Inconclusive
H59	At 1,000: fixed-shape bursts, fixed-only minus mixed corpora (two-sided; TOST $\pm$0.3)	−1.00	[−1.20, −0.80]	$<.001$		Lower in fixed-only corpora

H52: TOST p = 0.14; 90% interval [−0.33, −0.17]. H54: TOST $p <.001$. H55, pooled noun-level regression with size and seed effects and seed-clustered errors: 3.82 [3.18, 4.46]; without no-shape nouns: 3.66 [3.05, 4.28]. By size: 6.36, 2.47, 2.19.

Table J2: Experiment 3: copy score by corpus design, history and exposures (means over seeds by size, and seed-level mean with 95% interval, sizes averaged). Mixed: ten seeds; single-history designs: five seeds.

Corpus	History	Exp.	3.4M	10.9M	25.6M	Mean	95% CI
Mixed	Unseen	–	8.96	6.03	5.65	6.88	[6.31, 7.45]
	Fixed, spaced	100	1.10	1.53	2.10	1.58	[1.34, 1.82]
	Fixed, spaced	1,000	0.36	0.64	0.39	0.46	[0.27, 0.65]
	Fixed, bursts	100	6.17	3.76	4.36	4.76	[4.18, 5.35]
	Fixed, bursts	1,000	0.69	1.09	1.32	1.03	[0.83, 1.23]
	No shape, spaced	100	2.59	2.93	4.03	3.18	[2.68, 3.68]
	No shape, spaced	1,000	–0.02	0.92	1.67	0.86	[0.56, 1.15]
	No shape, bursts	100	3.16	4.66	5.27	4.36	[3.76, 4.96]
	No shape, bursts	1,000	0.06	1.55	1.54	1.05	[0.80, 1.30]
	Changing, spaced	100	1.63	2.12	2.17	1.97	[1.63, 2.32]
	Changing, spaced	1,000	0.28	0.63	0.79	0.57	[0.42, 0.71]
	Changing, bursts	100	9.03	4.53	4.77	6.11	[5.57, 6.66]
	Changing, bursts	1,000	7.05	3.53	3.56	4.71	[4.03, 5.40]
Changing only	Unseen	–	9.01	5.15	4.33	6.16	[4.25, 8.08]
	Changing, spaced	100	1.65	2.20	1.37	1.74	[0.96, 2.52]
	Changing, spaced	1,000	0.43	1.55	0.83	0.94	[0.44, 1.44]
	Changing, bursts	100	9.33	4.80	3.72	5.95	[4.09, 7.81]
	Changing, bursts	1,000	9.23	4.39	3.00	5.54	[4.12, 6.96]
Fixed only	Unseen	–	7.22	3.31	4.09	4.87	[3.84, 5.90]
	Fixed, spaced	100	0.18	1.05	1.26	0.83	[0.71, 0.95]
	Fixed, spaced	1,000	–0.16	0.11	0.08	0.01	[–0.10, 0.12]
	Fixed, bursts	100	1.68	1.55	2.09	1.77	[1.26, 2.29]
	Fixed, bursts	1,000	–0.23	0.31	0.00	0.03	[–0.13, 0.18]

Table J3: Experiment 3, mixed design, 1,000 exposures: copy score when the context sentence is about a noun of one history (rows) and the query names a different noun of another (columns); means over ten seeds, sizes averaged (24 items per pair and model).

Context \ query	Unseen	Fixed spaced	Fixed bursts	No shape spaced	No shape bursts	Changing spaced	Changing bursts
Unseen	6.87	0.67	0.97	0.99	1.16	0.51	2.98
Fixed, spaced	3.47	0.54	0.89	0.73	0.90	0.48	1.90
Fixed, bursts	5.23	0.71	1.13	0.94	1.16	0.60	3.02
No shape, spaced	4.12	0.40	0.85	0.83	1.00	0.41	1.82
No shape, bursts	4.84	0.61	0.94	0.87	0.87	0.51	2.71
Changing, spaced	4.08	0.47	0.91	0.93	1.08	0.56	2.16
Changing, bursts	7.09	0.81	1.28	1.32	1.44	0.80	4.73